

Tradeoffs Between Rate and Conditional Content with Encoder-Observed Context

Andrew H. Bond, *Senior Member, IEEE*

Abstract

An encoder observes a jointly Gaussian pair, a described variable and a correlated context, and describes the first for a decoder that sees the description alone. A third party retains a noisy copy of the context and never decodes. We study the conditional content of the stored description given that copy: the mutual information between source and reproduction conditioned on the third party's side information, Steinberg's common-reconstruction functional on the augmented source. Two results are primary. The conditional content has a closed form for jointly Gaussian sources, $L(D) = \frac{1}{2} \log_2 g^*$ with g^* the larger root of an explicit quadratic in the distortion level, the source–context correlation, and the noise in the third party's copy. And the region of jointly achievable rate and conditional content is exact, with a Pareto frontier traced by two coupled water levels; the two prices strictly trade off exactly when the context is partially informative, and the tradeoff persists when the copy is clean. A structural consequence is that the conditional content is not determined by the joint law of the described variable and the retained copy, so the encoder's access to the clean context enters the function. The closed form recovers the classical rate–distortion function, Gray's conditional function, and Steinberg's scalar Gaussian formula. The rate–content frontier of the doubly symmetric binary case is traced through a tilt equation, and coding theorems, including a direct proof for the Gaussian source, give the conditional content its operational meaning as the least normalized conditional entropy of the stored index.

Index Terms

Common reconstruction, conditional mutual information, Gaussian sources, rate–distortion theory, side information.

I. INTRODUCTION

A stored description of data has two prices. The first is its rate, the space the description occupies, and rate–distortion theory prices it against the fidelity the decoder requires. The second is paid at the end of the description's life: erasing a stored description irreversibly costs work, and when the mechanism performing the reset retains correlated information, the work is governed not by the description's entropy but by its conditional entropy given what the mechanism holds. The two prices are set by different parties. The decoder never sees the reset mechanism's information, and the reset mechanism never serves the decoder. This paper solves the source coding problem that this asymmetry creates: in closed form for Gaussian sources, and, for the doubly symmetric binary source, by tracing the rate–content frontier through a tilt equation.

The setting is as follows. An encoder observes a jointly Gaussian pair (Y, V) : Y is the variable the decoder must reproduce, and V is a correlated variable, which we call the context ($\rho = \mathbb{E}[YV]$ after normalization). The encoder emits a description under a distortion constraint $\mathbb{E}(Y - \hat{Y})^2 \leq D$ that binds the unconditional error; the decoder reconstructs from the description alone. A third party holds $S = V + U$, a noisy observation of the context with noise variance τ^2 . The quantity of interest is $L(D) = \min I(X; \hat{X} | S)$ over test channels satisfying the distortion constraint and the Markov chain $\hat{X} - X - S$. We call $L(D)$ the minimal conditional content of an admissible description; by the operational theorems of Section II and, for the Gaussian source, Appendix A, it is the least asymptotically achievable value of the normalized conditional entropy $H(M | S^n)/n$ of the stored index M among codes meeting the distortion constraint. The thermodynamic reading of this quantity, as the infimum of the average erasure work per symbol under an ideal conditional reset, is stated once, in Remark 8, together with the setting it needs.

A. H. Bond is with the Department of Computer Engineering, San José State University, San José, CA 95192 USA (e-mail: andrew.bond@sjsu.edu). Numerical verification harnesses and archived records: <https://github.com/ahb-sjsu/geometric-observation>; the manuscript sources and the verification scripts are archived at doi:10.5281/zenodo.22678341.

The single-letter functional itself is not new. With the augmented source $X = (Y, V)$ and a distortion that ignores the V coordinate, $\min I(X; \hat{X} | S)$ is exactly Steinberg's common-reconstruction functional [11], and Remark 8 of Lapidoth, Malär, and Wigger [12] notes precisely this augmentation device: an encoder observing an additional sequence correlated with the side information is handled by enlarging the source alphabet and letting the distortion ignore the extra coordinate. What this paper adds is the solution of the augmented problem, and two contributions are primary. First, the closed form: we evaluate the augmented Gaussian functional exactly, $L(D) = \frac{1}{2} \log_2 g^*$ with g^* the larger root of an explicit quadratic together with the achieving channel (Theorem 15(a)), after Lemma 11 reduces the problem, for Y and V scalar linear functionals of an arbitrary-dimensional jointly Gaussian source, to the pair (ρ, τ^2) . The known Gaussian solutions are scalar and unaugmented: Steinberg's Gaussian example [11] and the Gaussian theorem of [12] take a scalar source with side information $X + U$ and an encoder that observes the source alone, and the augmentation remark of [12] is stated for finite alphabets, so the augmented Gaussian functional is evaluated in neither. Second, the exact rate–content region with its strict misalignment dichotomy: the Pareto frontier is traced by a coupled two-water-level stationarity system, and within the strictly noisy model $\tau^2 > 0$ it is nondegenerate, so that rate and content strictly trade off, exactly when $0 < \rho^2 < 1$, with the tradeoff persisting at the clean-context boundary (Theorem 15(b) and Corollary 20). A structural consequence of the closed form is non-determination by the reduced marginal (Corollary 22): replacing the encoder's observation of the pair (Y, V) by its reduced relation to S changes the operational problem even when the reduced (Y, S) laws are equivalent up to invertible scaling. That is the one-line answer to why the function is not Steinberg's scalar formula: the encoder's access to the clean context V , which neither the decoder nor the third party shares, enters the function itself. The operational layer supports these results rather than leading them: coding theorems characterize $L(D)$ as the least asymptotically achievable normalized conditional entropy $H(M | S^n)/n$ of the stored index (Section II, and Appendix A for the Gaussian source), and the thermodynamic reading of $L(D)$ (Remark 8) is one consequence of that characterization.

The remaining results are extensions, consistency anchors, and one attainment remark. Remark 23 reads off the closed form when the conditional-variance lower bound is attained. Three extensions follow: the rate–content frontier of a doubly symmetric binary source with a binary symmetric context channel is traced through an elementary tilt equation anchored at Gray's binary conditional function and the marginal rate–distortion function (Section VII); the frontier extends to vector context (Section VI); and the operational region theorem is proved directly for the Gaussian source (Appendix A). The consistency anchors are the three classical corners of the closed form, Shannon's rate–distortion function at $\rho = 0$ [1], Gray's conditional function as $\tau^2 \rightarrow 0$ [4], and Steinberg's scalar Gaussian formula at the induced correlation as $\rho^2 \rightarrow 1$, together with the strict suboptimality of marginalization (Corollary 17) and an exact information-bottleneck identity for the weighted objective.

The dependency chain is short. Theorem 6 and Lemma 7 give the discrete region and the finite-blocklength construction; Lemma 11 reduces the Gaussian problem to the pair; Lemma 13 supplies the Gaussian exhaustion; Theorem 15(a) is the closed form; Proposition 19 and Theorem 15(b) give uniqueness and the region; Corollaries 20 and 22 follow; and Theorems 24, 27, and 29 are downstream.

Related work

Conditional rate–distortion theory is due to Gray [3], [4]: both terminals observe the conditioning variable, and $R_{Y|S}(D)$ prices the description under shared context. Here no terminal observes S , and Gray's function appears only as the clean-context limit $\tau^2 \rightarrow 0$ (Section IV). Wyner and Ziv [5] place the side information at the decoder, where it participates in reconstruction; our decoder holds nothing but the description, and Wyner–Ziv binning enters the operational theorem only as a proof device.

Steinberg [11] introduced the common-reconstruction constraint, under which the decoder holds side information but the encoder must be able to reproduce the decoder's output; the resulting functional $\min I(X; \hat{X} | S)$ under $\hat{X} = X - S$ is the single-letter object of this paper, and Steinberg solved its scalar Gaussian case. Lapidoth, Malär, and Wigger [12] relaxed exact to approximate reproduction and solved the scalar Gaussian problem with side information $X + U$; their Remark 8, stated for finite alphabets, incorporates an encoder observing an additional correlated sequence by augmenting the source and letting the distortion ignore the extra coordinate. Our problem is exactly that augmented problem, with $X = (Y, V)$ and distortion on Y alone. What is inherited is the functional and the augmentation device; what is new, to our knowledge, is the evaluation: the closed form and achieving channel

of Theorem 15(a) for Y and V scalar linear functionals of an arbitrary-dimensional source, the joint region with its two-water-level frontier (Theorem 15(b)) in place of the rate under a common-reconstruction constraint alone, the misalignment dichotomy (Corollary 20), and the non-determination corollary (Corollary 22). The binary conditional-content function coincides with the common-reconstruction rate of Lu *et al.* [21], and the binary contribution here is the joint frontier (Appendix C).

Side information held by the encoder alone is classically inert: when it does not enter the distortion measure, it leaves the rate–distortion function unchanged [2], and Kaspi [9] analyzed the configurations in which the encoder is informed of the side information. Here the encoder-held context V matters because the priced coordinate is conditional on S , with which V is correlated, and access to V strictly lowers the price for every $\rho \neq 0$ (Corollary 17 for $\tau^2 > 0$; Proposition 12 at the clean-context boundary). Heegard and Berger [10] serve decoders with and without side information from one description, and Ahmadi, Tandon, Simeone, and Poor [13] imposed the common-reconstruction constraint on the Heegard–Berger and cascade problems; Section VII distinguishes the binary frontier from their calculations.

The closest common-reconstruction counterpart is Lu and Xu’s vector Gaussian Heegard–Berger problem with common reconstructions [23]. Their Theorem 4 gives the rate–distortion function for two decoders with side information under error-covariance constraints $\text{Cov}(X - \hat{X}) \preceq \mathbf{D}_m$, with $\mathbf{D}_m$ diagonal and positive definite, under the hypothesis $\mathbf{D}_m \preceq \Sigma_X$ in the positive semidefinite order. Specialized to one decoder with side information S and constraint $\text{diag}(D, D_V)$ on the pair (Y, V) , with $s = 1 + \tau^2$, it reads $\frac{1}{2} \log_2 [(1 - \rho^2)(\tau^2 + D_V)/(sDD_V)]$: their Assumption 1 holds in the (Y, V) coordinates, since $\Sigma_{T|S}^{-1} - \Sigma_T^{-1} = \text{diag}(0, \tau^{-2})$, and the same specialization returns Steinberg’s scalar formula when the context is dropped. The present problem is the unpenalized-context limit $D_V \rightarrow \infty$ of the underlying Lu–Xu constraint family, at which the covariance constraint reduces to $\mathbb{E}(Y - \hat{Y})^2 \leq D$; every channel feasible for their constraint is feasible here, and dropping $\hat{V}$ lowers the conditional information by data processing, so their function is at least $L(D)$ wherever it is defined. What their closed form does not cover is that limit: its hypothesis $\mathbf{D} \preceq \Sigma_X$ forces $(1 - D)(1 - D_V) \geq \rho^2$, which leaves no admissible D_V unless $D < 1 - \rho^2$ and never leaves the context unpenalized. On the admissible range the inequality is strict: the specialized value decreases in D_V , and at the largest admissible D_V it equals $\frac{1}{2} \log_2 g_R$ with $P(g_R) = \rho^2 \tau^2 (1 - D)(1 - \rho^2)/(s(1 - D - \rho^2)^2) > 0$, so $g_R \neq g^*$ while $g_R \geq g^*$ by the inequality just proved. The formal continuation of the formula to $D_V = \text{Var } V$ equals Gray’s clean-context function $\frac{1}{2} \log_2 ((1 - \rho^2)/D)$, strictly below $L(D)$ whenever $\rho\tau \neq 0$, falling further beyond. Theorem 15(a) is therefore the value of a limit that their closed form is not stated for and does not continue to.

The priced coordinate also has a security reading: L is the equivocation $H(M | S^n)/n$ of the stored description at a party holding the noisy context. Yamamoto priced distortion against source equivocation in the Shannon cipher system [14] and, in an earlier problem [15], let the encoder observe a source together with an additional output whose equivocation at the receiver or wiretapper is to stay high; Villard and Piantanida [16] and Schieler and Cuff [17] developed secure source coding with side information at the eavesdropper and a rate–distortion theory for secrecy systems. That line prices the equivocation of the source and seeks to make it large, and its encoder does not observe the clean context; here the equivocation of the description is made small, as the cost of a conditional reset. Closer in form are the side-information-privacy problem of Tandon, Sankar, and Poor [18] and the privacy funnel of Makhdoumi, Salamati, Fawaz, and Médard [19], which price the leakage $I(\hat{X}; S)$ of the description about the side information. Under the Markov chain $I(X; \hat{X} | S) = I(X; \hat{X}) - I(\hat{X}; S)$, so the present problem rewards the leakage that the privacy line penalizes: the two share their coordinates with opposite signs.

The thermodynamic reading of L rests on Landauer’s principle [27] in its average-work form for arbitrary distributions [31] and with side information [33], [32], [34]; the single-shot sharpenings [29], [30] price the same reset by smooth entropies, and Remark 8 states the reading, its setting, and its scope.

The relation to the information bottleneck [26] is exact. For any channel with $\hat{Y} - (Y, V) - S$, writing $T = (Y, V)$, the weighted objective of Section V collapses by the Markov chain:

$$J_\alpha := \alpha I(T; \hat{Y}) + (1 - \alpha) I(T; \hat{Y} | S) = I(T; \hat{Y}) - (1 - \alpha) I(S; \hat{Y}), \quad (1)$$

an information-bottleneck Lagrangian with tradeoff parameter $1 - \alpha$, augmented by the hard distortion constraint $\mathbb{E}(Y - \hat{Y})^2 \leq D$. Gaussian IB [25] establishes the optimality of Gaussian linear-noise representations and the eigenmode structure of the unconstrained Lagrangian, and Meidlinger, Winkelbauer, and Matz [24] connected

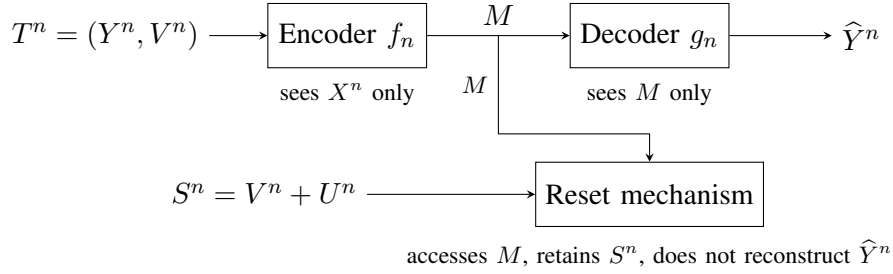


Fig. 1. The system. The encoder observes the source X^n alone and stores the description M at rate R_n ; the decoder reconstructs from M alone; the third party (the reset mechanism) retains the side information S^n and accesses the stored index M in order to erase it; it never reconstructs. The figure shows the Gaussian instantiation, where the encoder observes $T = (Y, V)$, $S = V + U$ with U independent of T , and the reproduction is $\hat{Y}^n$, an estimate of Y^n alone; the general discrete theorem reproduces the full letter.

it to mean-square rate–distortion quantization on the rate coordinate; the hard Y -distortion constraint binds a reconstruction error rather than a mutual information, and the two-water-level frontier is what it adds.

Finally, [35] is a preliminary manuscript by the author, archived at doi:10.5281/zenodo.21776291, where the discrete operational region theorem, its single-letterization, and the scalar Gaussian corner first appeared; everything from Section III onward is developed here, with the pair-sufficiency reduction credited to Armstrong’s source-side sufficiency [20] as a distortion-preserving specialization.

II. PROBLEM FORMULATION AND THE OPERATIONAL THEOREM

All logarithms are base two. Random variables have finite alphabets unless a Gaussian specialization is stated. The model has one source and three parties, and Fig. 1 shows it. Let $(X, S) \sim p_{XS}$, and let (X^n, S^n) be i.i.d. copies. In the Gaussian sections the encoder observes the pair $T = (Y, V)$, with Y the described variable and V the context, and the side information is $S = V + U$ with U independent of T ; there the source letter X is the ambient Gaussian vector of which T is the relevant pair (Lemma 11). We call S the *conditioner* when its role in the conditional content is in view (the terms side information and conditioner are synonyms for the variable, as third party and reset mechanism are for the party holding it). The operational theorem of this section needs no such structure. The encoder observes X^n alone and emits the description M . The decoder observes M alone and outputs the reproduction; the general discrete theorem below reproduces the full letter, $\hat{X}^n$, while in the Gaussian sections the reproduction is a scalar estimate $\hat{Y}$ of Y alone. The third party of Section I, called the *reset mechanism* when its erasure role is in view, retains S^n , which remains unchanged by the reset, and accesses the stored index M in order to erase it; it never decodes.

A distortion measure is a bounded function

$$d : \mathcal{X} \times \hat{\mathcal{X}} \longrightarrow [0, d_{\max}]. \quad (2)$$

An $(n, \mathcal{M}_n)$ observation code consists of an encoder and decoder

$$f_n : \mathcal{X}^n \rightarrow \mathcal{M}_n, \quad g_n : \mathcal{M}_n \rightarrow \hat{\mathcal{X}}^n. \quad (3)$$

We write $M = f_n(X^n)$ and $\hat{X}^n = g_n(M)$. Randomized encoders do not improve the region below, but allowing them in the converse is harmless.

The three operational quantities are

$$R_n = \frac{1}{n} \log |\mathcal{M}_n|, \quad (4)$$

$$D_n = \frac{1}{n} \sum_{i=1}^n \mathbb{E} d(X_i, \hat{X}_i), \quad (5)$$

$$L_n = \frac{1}{n} H(M | S^n). \quad (6)$$

The dimensionless quantity L_n is the residual description content: the uncertainty about the stored index that remains given the retained side information. We call it the *conditional content* of the memory relative to the retained side

information, a name introduced here for the normalized conditional entropy of the index; the thermodynamic reading of Remark 8, after the coding theorem, is the reason for pricing it. Every statement and proof of the paper treats L_n as the coding quantity (6), in bits per source symbol.

Definition 1. A triple (R, L, D) is achievable if there is a sequence of observation codes such that

$$\limsup_{n \rightarrow \infty} R_n \leq R, \quad \limsup_{n \rightarrow \infty} L_n \leq L, \quad \limsup_{n \rightarrow \infty} D_n \leq D. \quad (7)$$

The closure of the achievable set at distortion D is denoted $\mathcal{RW}(D)$.

For a test channel $p(\hat{x} | x)$, the Markov chain $\hat{X} - X - S$ holds because side information is unavailable when the observation is formed. Define

$$\mathcal{T}(D) = \left\{ p(\hat{x} | x) : \mathbb{E}d(X, \hat{X}) \leq D \right\}. \quad (8)$$

The two single-letter functions of the paper are the classical rate–distortion function and the content endpoint,

$$R_{\min}(D) = \min_{p(\hat{x}|x) \in \mathcal{T}(D)} I(X; \hat{X}), \quad L(D) = \min_{p(\hat{x}|x) \in \mathcal{T}(D)} I(X; \hat{X} | S). \quad (9)$$

Both minima exist by compactness of $\mathcal{T}(D)$ and continuity of the two coordinates. The coding theorem below gives both functions their operational meaning. The following box fixes the two notational scopes of the paper.

Definition 2. General letters, used for the discrete theorem of this section: source $(X, S) \sim p_{XS}$, i.i.d. copies (X^n, S^n) , test channel $p(\hat{x} | x)$ with the Markov chain $\hat{X} - X - S$, constraint set $\mathcal{T}(D)$, and the two functions of (9). Gaussian letters, used from Section III on: the source letter is the jointly Gaussian pair $T = (Y, V)$ with $\text{Var } Y = \text{Var } V = 1$ and $\mathbb{E}[YV] = \rho$; the context copy is generated as $S = V + U$ with $U \sim \mathcal{N}(0, \tau^2)$ independent of everything else; the test channel is $p(\hat{y} | t)$, the reproduction $\hat{Y}$ does not see S , and the Markov chain is $\hat{Y} - T - S$; the constraint is $\mathbb{E}(Y - \hat{Y})^2 \leq D$; and

$$R_{\min}(D) = \min I(T; \hat{Y}), \quad L(D) = \min I(T; \hat{Y} | S),$$

both minima over test channels meeting the constraint.

Lemma 3. For fixed p_{XS} , both coordinates $I(X; \hat{X})$ and $I(X; \hat{X} | S)$ are convex functions of the test channel $q(\hat{x} | x)$.

Proof. Convexity of $I(X; \hat{X})$ in the channel at fixed input is classical. For the conditional coordinate, the Markov chain $\hat{X} - X - S$ gives $p(\hat{x} | x, s) = q(\hat{x} | x)$, so

$$I(X; \hat{X} | S) = \sum_s p(s) I_{p(x|s)}(q), \quad (10)$$

a nonnegative combination of mutual informations, each with fixed input distribution $p(x | s)$ and the common channel q ; each term is convex in q , hence so is the sum. We note that the decomposition $I(X; \hat{X} | S) = I(X; \hat{X}) - I(S; \hat{X})$ does *not* yield this conclusion: $I(S; \hat{X})$ is itself convex in q (the induced channel $p(\hat{x} | s) = \sum_x p(x | s)q(\hat{x} | x)$ is linear in q), so the difference is a priori indefinite. $\square$

Remark 4. Lemma 3 holds verbatim for standard Borel alphabets, with the sums over s replaced by integrals against the law of S and the conditional laws furnished by regular conditional probabilities: the proof uses only the joint convexity of relative entropy [37] and the linearity of the induced output law in the channel. Theorem 15(b) and Theorem 29 use this form.

Lemma 5. For a finite-alphabet source (X, S) and bounded distortion d , define

$$\mathcal{R}(D) := \bigcup_{q \in \mathcal{T}(D)} \{ (R, L) : R \geq I_q(X; \hat{X}), L \geq I_q(X; \hat{X} | S) \}.$$

Then $\mathcal{R}(D)$ is closed and nondecreasing in D , and it is right-continuous:

$$\mathcal{R}(D) = \bigcap_{D' > D} \mathcal{R}(D').$$

Equivalently, if $(R_j, L_j) \in \mathcal{R}(D_j)$ with $(R_j, L_j) \rightarrow (R, L)$ and $\limsup_j D_j \leq D$, then $(R, L) \in \mathcal{R}(D)$.

Proof. Monotonicity is the nesting of the constraint sets $\mathcal{T}(D)$. For the sequential form, pick $q_j \in \mathcal{T}(D_j)$ with $R_j \geq I_{q_j}(X; \hat{X})$ and $L_j \geq I_{q_j}(X; \hat{X} | S)$. The channels live in the compact simplex of conditional distributions, so a subsequence converges to some q ; the distortion $\mathbb{E}_q d(X, \hat{X})$ is continuous in the channel, so $\mathbb{E}_q d = \lim_j \mathbb{E}_{q_j} d \leq \limsup_j D_j \leq D$ and $q \in \mathcal{T}(D)$. Both mutual-information coordinates are continuous in the channel on finite alphabets, so $I_q(X; \hat{X}) = \lim_j I_{q_j}(X; \hat{X}) \leq R$ and likewise $I_q(X; \hat{X} | S) \leq L$: the limit (R, L) lies in q 's quadrant, hence in $\mathcal{R}(D)$. Closedness is the sequential form with $D_j \equiv D$. For right-continuity, the inclusion $\mathcal{R}(D) \subseteq \bigcap_{D' > D} \mathcal{R}(D')$ is monotonicity, and conversely a point of the intersection is a constant sequence with $(R_j, L_j) \equiv (R, L) \in \mathcal{R}(D'_j)$ along any $D'_j \downarrow D$, so it lies in $\mathcal{R}(D)$ by the sequential form. $\square$

Architecture of the coding theorem: The proof of Theorem 6 below has six steps, and one possible confusion is worth preempting at the outset: the reconstruction decoder never sees S^n ; the binning of Wyner–Ziv type appears only inside an accounting argument. (i) The description rate comes from ordinary covering: a rate–distortion codebook whose index M the decoder alone uses. (ii) Reconstruction is from the full index M . (iii) A *virtual* bin index $B = b(M)$ is introduced for conditional-entropy accounting only; no party transmits or stores B . (iv) A Fano bound controls the residual uncertainty about M at a *hypothetical* decoder holding (B, S^n) ; this hypothetical decoder is a proof device, not a change of the model. (v) An expurgation step selects one deterministic code satisfying the covering and the accounting requirements simultaneously. (vi) The Gaussian source is treated in Appendix A, whose achievability invokes the finite-alphabet theorem below on a quantized surrogate; the closed forms of Sections III–VII are single-letter evaluations that do not depend on it.

Theorem 6. For a discrete memoryless source (X, S) and bounded distortion d ,

$$\mathcal{RW}(D) = \bigcup_{p(\hat{x}|x) \in \mathcal{T}(D)} \left\{ (R, L) : R \geq I(X; \hat{X}), L \geq I(X; \hat{X} | S) \right\}. \quad (11)$$

By Lemma 3 both coordinates are convex in the common test channel, so the union is already convex, and it is closed and right-continuous in the distortion level (Lemma 5); consequently no time sharing is required, and every Pareto-optimal point of the region is attained by a single test channel.

Proof. We first prove the converse. Let a code produce M and $\hat{X}^n$. Since $\hat{X}^n$ is a function of M ,

$$\begin{aligned} \log |\mathcal{M}_n| &\geq H(M) \geq I(X^n; M) \geq I(X^n; \hat{X}^n) \\ &\geq \sum_{i=1}^n I(X_i; \hat{X}_i), \end{aligned} \quad (12)$$

where the last step follows from memorylessness and conditioning reducing entropy (the standard single-letterization, as in [5]). Likewise,

$$\begin{aligned} H(M | S^n) &\geq I(X^n; M | S^n) \geq I(X^n; \hat{X}^n | S^n) \\ &\geq \sum_{i=1}^n I(X_i; \hat{X}_i | S_i). \end{aligned} \quad (13)$$

Each induced per-letter channel $p(\hat{x}_i | x_i)$ satisfies the Markov constraint $\hat{X}_i - X_i - S_i$: $\hat{X}_i$ is a function of $(X_i, X_{\neq i})$ and S_i is independent of $X_{\neq i}$ given X_i for i.i.d. pairs. Let $\bar{q}(\hat{x} | x) = \frac{1}{n} \sum_{i=1}^n p(\hat{X}_i = \hat{x} | X_i = x)$ be the mixture channel; the constraint survives the mixture because $p(s | x)$ is common across i . Its distortion is D_n , so $\bar{q} \in \mathcal{T}(D_n)$, and by Lemma 3,

$$R_n \geq \frac{1}{n} \sum_i I(X_i; \hat{X}_i) \geq I_{\bar{q}}(X; \hat{X}), \quad L_n \geq \frac{1}{n} \sum_i I(X_i; \hat{X}_i | S_i) \geq I_{\bar{q}}(X; \hat{X} | S),$$

which places (R_n, L_n) in the (already convex and closed) union of (11) at distortion D_n . The right side of (11) is the set $\mathcal{R}(D_n)$ of Lemma 5, so, for an achievable triple with $\limsup D_n \leq D$, the sequential form of that lemma places every limit point of (R_n, L_n) , a bounded sequence by Definition 1, and with it (R, L) , in the right side of (11) at D .

For achievability, fix $p(\hat{x} | x) \in \mathcal{T}(D)$ and $\epsilon > 0$. Generate a random codebook $\mathcal{C}$ of $2^{n(I(X;\hat{X})+2\epsilon)}$ independent $\hat{x}^n$ codewords drawn from the $\hat{X}$ -marginal, and, independently, a random binning b assigning each codeword index uniformly to one of $2^{n(I(X;\hat{X}|S)+3\epsilon)}$ bins. The covering encoder selects the smallest index among the codewords jointly (strongly) typical with x^n , declaring failure otherwise, and stores that index M ; the decoder reconstructs the codeword from M alone. Two failure events. First, E_{cov} : no jointly typical codeword exists; $\Pr(E_{\text{cov}}) \rightarrow 0$, averaged over $(X^n, \mathcal{C})$, by the covering lemma, and on E_{cov}^c the distortion is at most $D + \epsilon$ by typicality. Second, E_{bin} : the side-information proof device fails. Because $\hat{X} - X - S$,

$$I(X; \hat{X} | S) = I(X; \hat{X}) - I(S; \hat{X}), \quad (14)$$

so each bin holds $2^{n(I(S;\hat{X})-\epsilon)}$ codewords on average, a per-bin codeword rate ϵ below the packing threshold $I(S; \hat{X})$, and the Wyner–Ziv packing argument [5] (the Markov lemma for the typicality of the selected codeword with S^n , and the packing bound for wrong codewords in the bin) gives a decoder that recovers M from (B, S^n) , $B = b(M)$, with $\Pr(E_{\text{bin}}) \rightarrow 0$ averaged over $(X^n, S^n, \mathcal{C}, b)$: it outputs the unique codeword in bin B jointly typical with S^n and declares an error otherwise. If $I(S; \hat{X}) = 0$, so that $I(X; \hat{X} | S) = I(X; \hat{X})$, the number of bins exceeds the number of codewords and the bin index determines M with high probability; the conclusion is unchanged. This decoder is a proof device; the actual decoder of the code still receives no side information.

Selection of one deterministic code. Let $\epsilon_n := \mathbb{E}_{\mathcal{C}, b} [\Pr(E_{\text{cov}} | \mathcal{C}) + \Pr(E_{\text{bin}} | \mathcal{C}, b)] \rightarrow 0$. By Markov's inequality there is a deterministic pair $(\mathcal{C}, b)$ with $\Pr(E_{\text{cov}}) + \Pr(E_{\text{bin}}) \leq 2\epsilon_n$. Fix it. Its distortion satisfies $D_n \leq D + \epsilon + 2\epsilon_n d_{\text{max}}$, and by Fano's inequality applied to the proof device on this fixed code, $H(M | B, S^n) \leq 1 + 2\epsilon_n \log |\mathcal{M}_n| = 1 + 2\epsilon_n n(I(X; \hat{X}) + 2\epsilon)$, so

$$\begin{aligned} H(M | S^n) &\leq H(B) + H(M | B, S^n) \\ &\leq n(I(X; \hat{X} | S) + 3\epsilon) + 1 + 2\epsilon_n n(I(X; \hat{X}) + 2\epsilon) = n(I(X; \hat{X} | S) + 3\epsilon + o(1)). \end{aligned} \quad (15)$$

The selected deterministic code thus satisfies the distortion and the conditional-entropy requirements simultaneously: the stored index has ordinary description rate $I(X; \hat{X}) + 2\epsilon$, distortion at most $D + \epsilon + o(1)$, and conditional content at most $I(X; \hat{X} | S) + 3\epsilon + o(1)$. Letting $\epsilon \downarrow 0$ along a diagonal sequence of blocklengths exhibits the corner $(I(X; \hat{X}), I(X; \hat{X} | S))$ at distortion D ; since the union in (11) is already convex and closed (Lemmas 3 and 5), this exhibits the entire region. $\square$

The achievability construction delivers finite-blocklength guarantees that Appendix A invokes on a quantized surrogate; we record them as a lemma.

Lemma 7. *Let (X, S) be a discrete memoryless source with finite alphabets, d a distortion with maximum d_{max} , $p(\hat{x} | x)$ a test channel with $\hat{X} - X - S$ and finite reproduction alphabet, and $\epsilon > 0$. There is a sequence $\epsilon_n \rightarrow 0$, depending on the joint law, the channel, and ϵ , such that for all sufficiently large n there is a deterministic code with $|\mathcal{M}_n| \leq 2^{n(I(X;\hat{X})+2\epsilon)}$, distortion $D_n \leq \mathbb{E} d(X, \hat{X}) + \epsilon + 2\epsilon_n d_{\text{max}}$, and*

$$H(M | S^n) \leq n(I(X; \hat{X} | S) + 3\epsilon) + 1 + 2\epsilon_n n(I(X; \hat{X}) + 2\epsilon).$$

Proof. This is the selected code of the achievability part of the proof of Theorem 6, with the constants displayed at (15); ϵ_n is the averaged failure probability defined there. $\square$

The binning step in the proof has a simple interpretation. The full codeword index is required by the decoder. Given the side information, the codeword is identified up to a residual set of asymptotic size $2^{nI(X;\hat{X}|S)}$; only that residual uncertainty remains in the stored index.

Remark 8. *This remark states the paper's only physical claim; everywhere else L_n and $L(D)$ are the coding quantities (6) and (9). The reset model is that of stochastic thermodynamics with side information: the third party holds the record M and the classical side information S^n , which is read without disturbance and remains unchanged; it knows the joint law of (M, S^n) induced by the code; it may drive M by any protocol in contact with a heat bath at temperature T (the temperature, not the pair of Section III), conditioned on S^n ; the work supplied may fluctuate from run to run and only its mean is counted; and only M is required to return to a standard state. The memory is the ideal degenerate logical memory of Landauer's setting: its logical states carry no energy difference, the protocol is cyclic in the memory's Hamiltonian, and the side-information subsystem is left unchanged; the identity below is*

the infimum in that model, not a claim about a particular device. Under these conditions the generalized second law with correlations [32], [34] bounds the average work of any process that resets M below by $k_B T \ln 2 H(M | S^n)$: Landauer's bound [27], valid for arbitrary distributions [31], with the entropy of the record reduced by its mutual information with the retained side information. The bound is approached, though not reached in finite time, by quasi-static isothermal protocols conditioned on S^n [31], [33], so the infimum of the average work over protocols is

$$W_n^* = k_B T \ln 2 H(M | S^n) = n k_B T \ln 2 L_n, \quad (16)$$

for every code. Hence, by Theorem 6, and by Theorem 29 for the Gaussian source, $L(D)$ is the infimum of the asymptotic average erasure work per symbol, in units of $k_B T \ln 2$, over codes meeting the distortion constraint. This is the average-work setting. The single-shot, deterministic-work setting of [29], [30] prices the same reset by smooth conditional entropies of the pair (M, S^n) , and we make no claim in that setting. The scope of the claim: (16) prices only the logically irreversible reset of the volatile index M ; it does not count the work of acquiring X^n , driving a channel, correcting errors, or finishing in finite time [27], [28], and the codebook, wiring, and any permanent lookup tables are part of the apparatus and are not erased in each cycle. Physical units are restored by multiplying any content value in bits per symbol by $k_B T \ln 2$.

The content-only endpoint of the region is the minimum conditional content, the function $L(D)$ defined in (9). As an information quantity, $L(D)$ is Steinberg's common-reconstruction rate–distortion function for side information S [11]: the reproduction must be determined without access to S . The operational roles differ. In the common-reconstruction problem, S is decoder side information that must not corrupt reproducibility; here S is held by a reset mechanism that never serves the decoder. The remainder of the paper characterizes $L(D)$, and the region around it, for the source of Section III.

Remark 9. From Section III on, the source letter is the jointly Gaussian pair $T = (Y, V)$ introduced there (the ambient source X of that section enters only through T , by Lemma 11), and the distortion is $d(x, \hat{x}) = (y - \hat{y})^2$ on the Y coordinate: the decoder is judged on Y alone, while the encoder observes the pair.

Remark 10. Theorem 6 is stated and proved for finite alphabets and bounded per-letter distortion. For the jointly Gaussian source of Section III with quadratic distortion on the described variable, the corresponding operational statement is Theorem 29 in Appendix A: the converse is proved there in full, and the achievability by quantizing the conditioner, the source, and the reproduction, invoking the finite-alphabet theorem, and passing to the limit. The Gaussian closed forms of Sections III–VI are evaluations of the single-letter coordinates in (11), and their operational meaning is inherited through Theorem 29.

Table I collects the notation used from here on.

III. PAIR SUFFICIENCY

Let $X \sim \mathcal{N}(0, \Sigma_X)$ on $\mathbb{R}^d$, i.i.d. across source symbols. The described variable is the linear functional $Y = w^\top X$; the third party's copy is a second linear functional observed in noise, $S = \gamma^\top X + U$. Normalize $\text{Var}(w^\top X) = 1$ (rescaling D), write $V = \gamma^\top X / \sigma_\gamma$, $\sigma_\gamma^2 := \gamma^\top \Sigma_X \gamma$, the normalized context, with $\rho := \mathbb{E}[YV]$, $|\rho| < 1$, and rescale S (invariantly for all mutual informations) to $S = V + U$, $U \sim \mathcal{N}(0, \tau^2)$, $\tau^2 > 0$. Write $s = 1 + \tau^2$, the context-noise parameter (a scalar, not to be confused with the random variable S). Throughout, X , U , and all channel randomness (including any resampling randomness below) are *mutually* independent. This joint form of the independence assumption is essential: it is exactly what the S -kernel argument in the next proof uses.

From here on the Gaussian scope of Definition 2 is in force: the ambient source is X , the pair is $T = (Y, V)$, the reproduction is $\hat{Y}$, and once Lemma 11 below is proved, all information quantities are written on $(T, \hat{Y}, S)$. Reproductions may be taken scalar without loss of optimality: replacing $\hat{X}$ by $\hat{Y} := w^\top \hat{X}$ leaves the distortion unchanged and can only decrease $I(X; \cdot | S)$ (conditional data processing), and $\hat{Y} - X - S$ persists.

Standing assumptions and degeneracies: The Gaussian sections run under $0 < D < 1$, $|\rho| < 1$, and $\tau^2 > 0$, and every boundary is handled where it arises rather than excluded silently. For $D \geq 1$ the zero reproduction gives $R = L = 0$, and $D = 0$ is the extended-real limit; both are recorded with the Gaussian operational theorem (Appendix A). The boundaries $\rho^2 \in \{0, 1\}$ are the anchor corners, the classical function at $\rho = 0$ and Steinberg's scalar formula as $\rho^2 \rightarrow 1$ (Corollary 16), and the frontier collapse at both is computed in Corollary 20. The

TABLE I
NOTATION.

Symbol	Meaning
$T = (Y, V)$	encoder-observed pair: described variable Y , context V (normalized in Section III); X is the ambient source, the source letter of the discrete theorem
$S = V + U$	the third party's copy; $U \sim \mathcal{N}(0, \tau^2)$ independent of X
$s = 1 + \tau^2$	the context-noise parameter, the variance of S after normalization (scalar conditioner; the vector-context section uses Σ_U , with conditioner covariance $\Sigma_S = \Sigma_V + \Sigma_U$)
$\rho = \mathbb{E}[YV]$	correlation of the described variable and the context
D	distortion level, $\mathbb{E}(Y - \hat{Y})^2 \leq D$
M, R_n, L_n	stored index; rate $\frac{1}{n} \log \mathcal{M}_n $; conditional content $\frac{1}{n} H(M S^n)$
$R_{\min}(D), L(D)$	the two single-letter functions of (9)
<i>Derived quantities</i>	
$(a, b), \nu$	regression coordinates of the test channel $\hat{Y} = aY + bV + N$, and the noise variance $\nu = \text{Var } N$ (Section IV)
Q_0, Q_1	unconditional and S -conditional signal parts of the reproduction, $\text{Var}(aY + bV)$ and $\text{Var}(aY + bV S)$ (Section IV)
$h(a, b)$	noise-free distortion contribution $\text{Var}((1-a)Y - bV)$, so that $\mathbb{E}(Y - \hat{Y})^2 = h + \nu$ (25)
$\mu_c = (a\rho + b)/s$	regression coefficient of the reproduction on S (proof of Theorem 15)
$P(g), g^*$	the quadratic (20) and its larger root; $L(D) = \frac{1}{2} \log_2 g^*$
$\alpha; \gamma_0, \gamma_1$	frontier weight (rate versus content); the two water levels $\nu/(Q_0 + \nu)$ and $\nu/(Q_1 + \nu)$ (Theorem 15(b))
$L_R^*(D), R_L^*(D)$	conditional content of the rate-optimal channel; rate of the content-optimal channel (Corollary 20)
Λ, c, y_0	whitened conditional covariance, regression vector, and the unit vector with $Y = y_0^\top Z$, $Z = WT$ the whitened pair of (37) (Sections V, VI)

clean-context boundary $\tau^2 = 0$ has its own branch wherever it matters: Proposition 12 and the $\tau^2 \rightarrow 0$ anchor for the function, and Proposition 21 for the frontier; positivity $\tau^2 > 0$ is exactly what makes $\text{Cov}(T, S) \succ 0$, the hypothesis of the exhaustion lemma (Lemma 13). Degenerate covariances reduce: $\Sigma_T \succ 0$ is assumed, with linearly dependent components dropped where noted, and singular channel residuals Σ_N are dispatched by Remark 14. The closed form's root satisfies $g^* > 1$ throughout $0 < D < 1$, $\tau^2 > 0$, by the bracketing $P(1) = (D - 1)\tau^2 < 0$ (Theorem 15).

Lemma 11. *In the problem $L(D) = \min\{I(X; \hat{Y} | S) : \hat{Y} - X - S, \mathbb{E}(Y - \hat{Y})^2 \leq D\}$, it is without loss of optimality to let the channel depend on X only through $T = (Y, V)$, and then $I(X; \hat{Y} | S) = I(T; \hat{Y} | S)$. Consequently $L(D)$ depends on $(\Sigma_X, w, \gamma, \tau^2)$, for the scalar linear functionals $Y = w^\top X$ and $V = \gamma^\top X / \sigma_\gamma$ of a source of arbitrary dimension d , only through (ρ, τ^2) .*

Proof. Given an admissible $p(\hat{y} | x)$, draw $\hat{Y}'$ from the conditional law $p(\hat{y} | T = t)$ using fresh randomness. (a) $(T, \hat{Y}') \stackrel{d}{=} (T, \hat{Y})$, so the distortion (a functional of $(Y, \hat{Y})$) is unchanged. (b) $S = V + U$ with U independent of $(X, \hat{Y})$ and of $(X, \hat{Y}')$, so in both joints the conditional density of S given $(T, \hat{Y}) = (t, \hat{y})$ is $p_U(\cdot - v)$: hence $(T, \hat{Y}', S) \stackrel{d}{=} (T, \hat{Y}, S)$. (c) Given T , $\hat{Y}'$ is independent of (X, S) , so $I(X; \hat{Y}' | S) = I(T; \hat{Y}' | S) + I(X; \hat{Y}' | T, S) = I(T; \hat{Y} | S) \leq I(X; \hat{Y} | S)$, the last step because T is a function of X . (d) $\hat{Y}' - X - S$ holds since $\hat{Y}'$ is a function of $(X, \text{fresh noise})$ with the noise independent of (X, U) . The same computation applied to a T -channel directly gives $I(X; \hat{Y} | S) = I(T; \hat{Y} | S)$. $\square$

This reduction is a distortion-preserving specialization of source-side sufficiency for the information bottleneck. Armstrong [20] establishes that replacing the source by a deterministic statistic sufficient for the relevance variable leaves the bottleneck tradeoff unchanged at every blocklength; Lemma 11 is the case in which the source is X and the statistic is $T = (Y, V)$, with the addition that the T -measurable mean-square distortion constraint is preserved, so the reduction holds for the constrained functional and not only the unconstrained bottleneck.

IV. THE FUNCTION: SCALAR CONTEXT

One further reduction identifies the class the closed form must beat. (*Marginalized class = scalar corner.*) If the channel depends on X only through Y (that is, $\hat{Y} - Y - (X, S)$), then given (Y, S) , $\hat{Y}$ is independent of X ,

so $I(X; \hat{Y} | S) = I(Y; \hat{Y} | S)$, and the class optimum is exactly the scalar-corner value $\frac{1}{2} \log_2 [(\sigma^2(1 - \rho_{YS}^2) + \rho_{YS}^2 D)/D]$ of the pair (Y, S) , where $\sigma^2 = \text{Var } Y$ and ρ_{YS} is the correlation of that pair, $\rho_{YS}^2 = \rho^2/s$ after the normalization of Section III [11]; the corner's converse applies to every $p(\hat{y} | y)$, discrete or not.

Proposition 12. *Let $d = 2$, $X \sim \mathcal{N}(0, I_2)$, $Y = X_1$, and $S = X_1 + X_2$ (so $\rho_{YS}^2 = \frac{1}{2}$ and S is X -measurable). For $0 < D < 1$:*

- (i) *the marginalized-class optimum is the scalar corner $L_{\text{marg}}(D) = \frac{1}{2} \log_2 \frac{1+D}{2D}$;*
- (ii) *the vector problem has the exact value*

$$L_{\text{vec}}(D) = \frac{1}{2} \log_2^+ \frac{1}{2D} = R_{Y|S}(D),$$

Gray's conditional rate-distortion function of (Y, S) [4] ($\text{Var}(Y | S) = \frac{1}{2}$), attained by the linear channel $\hat{Y} = (1 - D)X_1 + DX_2 + N$, $N \sim \mathcal{N}(0, D(1 - 2D))$, for $D \leq \frac{1}{2}$, and by $\hat{Y} = S/2$ (zero conditional content) for $D \geq \frac{1}{2}$;

- (iii) *hence the strict gap $L_{\text{marg}}(D) - L_{\text{vec}}(D) = \frac{1}{2} \log_2(1 + D) > 0$ on $(0, \frac{1}{2}]$, and equals the full marginalized value on $[\frac{1}{2}, 1)$ (where $L_{\text{vec}} = 0$; the absolute gap is decreasing there, but it is all of L_{marg}).*

Proof. (i) is the reduction above, evaluated at $\sigma^2 = 1$, $\rho_{YS}^2 = \frac{1}{2}$.

(ii) *Converse.* For any admissible $\hat{Y}$ (Markov $\hat{Y} - X - S$), $I(X; \hat{Y} | S) \geq I(Y; \hat{Y} | S)$, and the induced conditional channel $p(\hat{y} | y, s)$ is feasible for Gray's conditional problem at the same distortion, so $I(Y; \hat{Y} | S) \geq R_{Y|S}(D) = \frac{1}{2} \log_2^+(\text{Var}(Y | S)/D) = \frac{1}{2} \log_2^+(1/(2D))$. *Achievability.* For $D \leq \frac{1}{2}$ take $\hat{Y} = (1 - D)X_1 + DX_2 + N$: distortion $D^2 + D^2 + D(1 - 2D) = D$; $\text{Var}(\hat{Y}) = 1 - D$, $\text{Cov}(\hat{Y}, S) = 1$, $\text{Var}(\hat{Y} | S) = 1 - D - \frac{1}{2} = \frac{1}{2} - D$, so $I(X; \hat{Y} | S) = \frac{1}{2} \log_2 \frac{(1-2D)/2}{D(1-2D)} = \frac{1}{2} \log_2 \frac{1}{2D}$. For $D \geq \frac{1}{2}$, $\hat{Y} = S/2$ is a deterministic function of X , has distortion $\mathbb{E}((X_1 - X_2)/2)^2 = \frac{1}{2} \leq D$, and is constant given S , so $I(X; \hat{Y} | S) = 0$.

(iii) Subtract. □

The closed form occupies the rest of this section. By Lemma 11 we restrict to T -channels, $T = (Y, V)$, with scalar reproduction $\hat{Y}$. The working coordinates deserve a plain statement before the machinery. Every jointly Gaussian test channel takes the form $\hat{Y} = aY + bV + N$ with N independent noise of variance ν , and every admissible channel, Gaussian or not, is measured through the same three moments of its regression on T : the unconditional signal part $Q_0 = \text{Var}(aY + bV)$, the S -conditional signal part $Q_1 = \text{Var}(aY + bV | S)$, and the noise-free distortion contribution $h = \text{Var}((1 - a)Y - bV)$, so that $\mathbb{E}(Y - \hat{Y})^2 = h + \nu$. Everything from here through Section VI is expressed in (a, b, ν) and these moments; Table I collects them. The converse rests on one estimation-theoretic lemma. We state it for vector reproductions and vector conditioners, in the generality that Sections V and VI reuse.

Lemma 13. *Let (T, S) be jointly Gaussian and centered, $T \in \mathbb{R}^p$, $S \in \mathbb{R}^r$, with $\text{Cov}((T, S)) \succ 0$. Let $\hat{\mathbf{Y}} \in \mathbb{R}^m$ be any centered random vector with finite second moments, and define*

$$A := \text{Cov}(\hat{\mathbf{Y}}, T) \Sigma_T^{-1} \in \mathbb{R}^{m \times p}, \quad N' := \hat{\mathbf{Y}} - AT, \quad \Sigma_N := \text{Cov}(N') = \text{Cov}(\hat{\mathbf{Y}}) - A \Sigma_T A^\top.$$

Assume $\Sigma_N \succ 0$ and $\text{Cov}(N', S) = 0$. Then

$$I(T; \hat{\mathbf{Y}} | S) \geq \frac{1}{2} \log_2 \frac{\det \Sigma_{T|S}}{\det \Sigma_e} = \frac{1}{2} \log_2 \frac{\det(A \Sigma_{T|S} A^\top + \Sigma_N)}{\det \Sigma_N}, \quad (17)$$

where $\Sigma_{T|S} = \Sigma_T - \Sigma_{TS} \Sigma_S^{-1} \Sigma_{ST}$ and Σ_e is the error covariance of the best linear estimator of T from $(\hat{\mathbf{Y}}, S)$. With S deleted throughout, the hypothesis $\text{Cov}(N', S) = 0$ becomes vacuous and the same argument gives

$$I(T; \hat{\mathbf{Y}}) \geq \frac{1}{2} \log_2 \frac{\det(A \Sigma_T A^\top + \Sigma_N)}{\det \Sigma_N}. \quad (18)$$

The jointly Gaussian channel $\hat{\mathbf{Y}} = AT + N$, $N \sim \mathcal{N}(0, \Sigma_N)$ independent of (T, S) , realizes the same second moments and attains both bounds with equality simultaneously.

Proof. Step 1: the max-entropy bound. The identity $I(T; \hat{\mathbf{Y}} | S) = h(T | S) - h(T | \hat{\mathbf{Y}}, S)$ is used as an extended-real identity: mutual information is the relative entropy of the joint law with respect to the product of its

marginals [6], and $h(T | \hat{Y}, S) = -\infty$ is the case of infinite conditional information that Remark 14 treats. Write $M := \text{Cov}((\hat{Y}, S))$ and $J := \text{Cov}(T, (\hat{Y}, S)) \in \mathbb{R}^{p \times (m+r)}$. First, $M \succ 0$: by the Schur-complement criterion on

$$M = \begin{bmatrix} \text{Cov}(\hat{Y}) & \text{Cov}(\hat{Y}, S) \\ \text{Cov}(S, \hat{Y}) & \Sigma_S \end{bmatrix},$$

positivity is equivalent to $\Sigma_S \succ 0$ together with $\text{Cov}(\hat{Y}) - \text{Cov}(\hat{Y}, S)\Sigma_S^{-1}\text{Cov}(S, \hat{Y}) \succ 0$; the first holds because $\text{Cov}(T, S) \succ 0$, and the second is evaluated in Step 2 to equal $A\Sigma_{T|S}A^\top + \Sigma_N \succeq \Sigma_N \succ 0$. The best linear estimator of T from $(\hat{Y}, S)$ is $\hat{T} = JM^{-1}(\hat{Y}; S)$, with error covariance $\Sigma_e = \Sigma_T - JM^{-1}J^\top$. Since (T, S) is jointly Gaussian, $h(T | S) = \frac{1}{2} \log_2((2\pi e)^p \det \Sigma_{T|S})$ exactly. For the other term, translation invariance of differential entropy, the fact that conditioning reduces entropy, and the Gaussian maximum-entropy bound at fixed covariance give

$$h(T | \hat{Y}, S) = h(T - \hat{T} | \hat{Y}, S) \leq h(T - \hat{T}) \leq \frac{1}{2} \log_2((2\pi e)^p \det \Sigma_e)$$

(with the usual convention that $h = -\infty$ for laws without a density, in which case the bound is trivial). Subtracting the two displays inside $I(T; \hat{Y} | S) = h(T | S) - h(T | \hat{Y}, S)$ yields the inequality in (17).

Step 2: the determinant identity. Let $K := \text{Cov}((T, S, \hat{Y}))$. The residual N' is uncorrelated with T by construction of the regression and with S by hypothesis, and the change of variables $(T, S, \hat{Y}) \mapsto (T, S, N')$ is block lower triangular with unit diagonal, hence determinant preserving:

$$\det K = \det \text{Cov}((T, S, N')) = \det \begin{bmatrix} \Sigma_T & \Sigma_{TS} & 0 \\ \Sigma_{ST} & \Sigma_S & 0 \\ 0 & 0 & \Sigma_N \end{bmatrix} = \det \Sigma_S \det \Sigma_{T|S} \det \Sigma_N,$$

using in the last step the Schur factorization $\det \text{Cov}(T, S) = \det \Sigma_S \det \Sigma_{T|S}$. The same Schur factorization applied to K in the other block order gives $\det K = \det M \det \Sigma_e$, and applied to M itself gives

$$\det M = \det \Sigma_S \cdot \det \left(\text{Cov}(\hat{Y}) - \text{Cov}(\hat{Y}, S)\Sigma_S^{-1}\text{Cov}(S, \hat{Y}) \right).$$

It remains to evaluate the conditional block. From $\hat{Y} = AT + N'$ with $\text{Cov}(N', T) = 0$ and $\text{Cov}(N', S) = 0$, $\text{Cov}(\hat{Y}) = A\Sigma_T A^\top + \Sigma_N$ and $\text{Cov}(\hat{Y}, S) = A\Sigma_{TS}$, so

$$\text{Cov}(\hat{Y}) - \text{Cov}(\hat{Y}, S)\Sigma_S^{-1}\text{Cov}(S, \hat{Y}) = A(\Sigma_T - \Sigma_{TS}\Sigma_S^{-1}\Sigma_{ST})A^\top + \Sigma_N = A\Sigma_{T|S}A^\top + \Sigma_N.$$

Combining the three displays,

$$\det \Sigma_e = \frac{\det K}{\det M} = \frac{\det \Sigma_{T|S} \det \Sigma_N}{\det (A\Sigma_{T|S}A^\top + \Sigma_N)},$$

which is the identity in (17); in particular $\Sigma_e \succ 0$, so Step 1's bound is finite. Deleting S (formally $r = 0$: $K = \text{Cov}(T, \hat{Y})$, $M = \text{Cov}(\hat{Y})$) gives $\det \Sigma_{e0} = \det \Sigma_T \det \Sigma_N / \det (A\Sigma_T A^\top + \Sigma_N)$ and hence (18).

Step 3: equality. For the stated Gaussian channel the conditional law of T given $(\hat{Y}, S)$ is Gaussian with covariance Σ_e for every realization, so $h(T | \hat{Y}, S) = \frac{1}{2} \log_2((2\pi e)^p \det \Sigma_e)$ and (17) holds with equality; likewise for (18). The channel's moments match those of the given channel: $\text{Cov}(AT + N, T) = A\Sigma_T$ and $\text{Cov}(AT + N) = A\Sigma_T A^\top + \Sigma_N$. $\square$

Remark 14. If Σ_N is singular, some combination $\eta^\top \hat{Y}$ equals $\eta^\top AT$ almost surely. If $\eta^\top A \neq 0$ the reproduction reveals a noiseless linear functional of the continuous T , so $I(T; \hat{Y} | S) = \infty$ and every bound below holds trivially; if $\eta^\top A = 0$ that combination of the reproduction is zero and may be dropped, reducing m . All uses of Lemma 13 below may therefore assume $\Sigma_N \succ 0$.

Theorem 15. With $s = 1 + \tau^2$, the context-noise parameter of Section III, for $0 < D < 1$:

(a) The closed form.

$$L(D) = \frac{1}{2} \log_2 g^*, \quad g^* = \frac{(D + s - \rho^2) + \sqrt{(D + s - \rho^2)^2 - 4Ds(1 - \rho^2)}}{2Ds}, \quad (19)$$

the larger root of the quadratic

$$P(g) = Dsg^2 - (D + s - \rho^2)g + (1 - \rho^2). \quad (20)$$

Since $P(1) = (D-1)\tau^2 < 0$, exactly one root exceeds 1, so g^* is well defined. The optimum is attained by the jointly Gaussian channel

$$\hat{Y} = aY + bV + N, \quad a = \frac{g^* - 1}{g^*}, \quad b = \frac{(g^* - 1)\rho}{g^*k}, \quad k = g^*s - 1, \quad N \sim \mathcal{N}\left(0, \frac{Q_1}{g^* - 1}\right), \quad (21)$$

where $Q_1 = a^2 + b^2 + 2ab\rho - (a\rho + b)^2/s$ is evaluated at these (a, b) .

(b) The region. The union of the quadrants above the single-letter rate-content pairs $(I(T; \hat{Y}), I(T; \hat{Y} | S))$ over admissible test channels at distortion D , which Theorem 29 identifies with the operational region $\mathcal{RW}(D)$, is the closed convex set whose Pareto frontier is traced by $\alpha \in [0, 1]$ as follows. Let $(a, b, \nu, \gamma_0, \gamma_1)$, with $\nu > 0$ and $\gamma_0, \gamma_1 \in (0, 1)$, solve the stationarity system

$$a = 1 - \alpha\gamma_0 - (1 - \alpha)\gamma_1, \quad \mu_c = \frac{a\rho}{s - (1 - \alpha)\gamma_1}, \quad b = (1 - \alpha)\gamma_1\mu_c, \quad (22)$$

$$\nu = D - h(a, b), \quad \gamma_0 = \frac{\nu}{Q_0 + \nu}, \quad \gamma_1 = \frac{\nu}{Q_1 + \nu}. \quad (23)$$

The solution exists and is unique at every α (Proposition 19 in Section V; at $\alpha = 0$ uniqueness also follows from part (a)), and the frontier point is

$$(R(\alpha), L(\alpha)) = \left(\frac{1}{2} \log_2 \frac{1}{\gamma_0}, \frac{1}{2} \log_2 \frac{1}{\gamma_1}\right),$$

achieved by $\hat{Y} = aY + bV + N$, $N \sim \mathcal{N}(0, \nu)$. Endpoints: $\alpha = 1$ recovers the classical reverse channel ($b = 0$, $a = 1 - D$, $\gamma_0 = D$, $R = \frac{1}{2} \log_2(1/D)$); $\alpha = 0$ recovers the channel and value of part (a) ($\gamma_1 = 1/g^*$).

Part (a) is proved next. Part (b) is proved in Section V, after Proposition 19 there supplies the uniqueness it uses.

Proof of part (a). Moments. By Lemma 11 and the scalar-reproduction reduction of Section III, restrict to channels $p(\hat{y} | t)$. Fix an admissible channel with $\mathbb{E}(Y - \hat{Y})^2 \leq D$. Centering $\hat{Y}$ leaves $I(T; \hat{Y} | S)$ unchanged and can only decrease the distortion, so take $\mathbb{E}\hat{Y} = 0$. Define the regression coordinates

$$\begin{pmatrix} a \\ b \end{pmatrix} := \Sigma_T^{-1} \begin{pmatrix} \mathbb{E}[Y\hat{Y}] \\ \mathbb{E}[V\hat{Y}] \end{pmatrix}, \quad Q_0 := a^2 + b^2 + 2ab\rho = \text{Var}(aY + bV), \quad \nu := \text{Var}\hat{Y} - Q_0,$$

so that $\mathbb{E}[Y\hat{Y}] = a + b\rho$ and $\mathbb{E}[V\hat{Y}] = a\rho + b$; $\nu \geq 0$ is the residual variance of $\hat{Y}$ after linear regression on T . If $\nu = 0$, Remark 14 applies: either $(a, b) \neq (0, 0)$ and $I(T; \hat{Y} | S) = \infty$, or $(a, b) = (0, 0)$ and $\hat{Y} = 0$ with distortion $1 > D$. So take $\nu > 0$.

The Markov moment identity. The residual $N' = \hat{Y} - aY - bV$ is uncorrelated with T by construction. It is also uncorrelated with $S = V + U$: it is uncorrelated with V , a coordinate of T , and $\mathbb{E}[UN'] = 0$ because U is independent of $(X, \hat{Y})$ by the mutual independence of X , U , and the channel randomness. Hence $\text{Cov}(N', S) = 0$, which is the hypothesis of Lemma 13, and which restates the moment identity $\mathbb{E}[S\hat{Y}] = \mathbb{E}[V\hat{Y}]$.

The converse bound, written out. Here $p = 2$, $m = 1$, $r = 1$, $A = (a, b)$, $\Sigma_N = \nu$, and

$$\Sigma_T = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}, \quad \Sigma_{TS} = \begin{bmatrix} \rho \\ 1 \end{bmatrix}, \quad \Sigma_S = s, \\ \Sigma_{T|S} = \Sigma_T - \frac{1}{s} \Sigma_{TS} \Sigma_{TS}^\top = \begin{bmatrix} 1 - \rho^2/s & \rho\tau^2/s \\ \rho\tau^2/s & \tau^2/s \end{bmatrix}, \quad \det \Sigma_{T|S} = \frac{\tau^2(1 - \rho^2)}{s} > 0.$$

The quadratic form in (17) is

$$A \Sigma_{T|S} A^\top = Q_0 - \frac{(a\rho + b)^2}{s} = \text{Var}(aY + bV | S) =: Q_1,$$

and, explicitly, $M = \text{Cov}((\hat{Y}, S)) = \begin{bmatrix} Q_0 + \nu & a\rho + b \\ a\rho + b & s \end{bmatrix}$ with $\det M = s(Q_0 + \nu) - (a\rho + b)^2 = s(Q_1 + \nu)$, while $\det \text{Cov}((T, S, \hat{Y})) = s \det \Sigma_{T|S} \cdot \nu$, so $\det \Sigma_e = \nu \det \Sigma_{T|S} / (Q_1 + \nu)$ and Lemma 13 reads

$$I(T; \hat{Y} | S) \geq \frac{1}{2} \log_2 \frac{\det \Sigma_{T|S}}{\det \Sigma_e} = \frac{1}{2} \log_2 \frac{Q_1 + \nu}{\nu}. \quad (24)$$

The distortion is a function of the same moments:

$$\mathbb{E}(Y - \hat{Y})^2 = 1 - 2(a + b\rho) + Q_0 + \nu = h(a, b) + \nu, \quad h(a, b) := (1 - a)^2 - 2(1 - a)b\rho + b^2, \quad (25)$$

and $h(a, b) = \text{Var}((1 - a)Y - bV) \geq 0$. Therefore

$$L(D) \geq \inf \left\{ B(a, b, \nu) := \frac{1}{2} \log_2 \frac{Q_1 + \nu}{\nu} : h(a, b) + \nu \leq D, \nu > 0 \right\}, \quad (26)$$

and it suffices to evaluate the right side; achievability below matches it.

Activity, coercivity, existence. At fixed (a, b) with $Q_1 > 0$, B is strictly decreasing in ν ($\partial_\nu [\ln(Q_1 + \nu) - \ln \nu] = (Q_1 + \nu)^{-1} - \nu^{-1} < 0$), so the infimum is unchanged by imposing the active constraint $\nu = D - h(a, b)$. On the feasible set, (a, b) is confined to the compact set $\{h \leq D\}$; since $\Sigma_{T|S} \succ 0$, the form $Q_1 = (a, b) \Sigma_{T|S} (a, b)^\top$ vanishes only at $(a, b) = (0, 0)$, where $h = 1 > D$, so by continuity $Q_1 \geq q_0 > 0$ on the feasible set. Hence $B \geq \frac{1}{2} \log_2((q_0 + \nu)/\nu) \rightarrow \infty$ as $\nu \rightarrow 0$, the sublevel sets of the reduced objective

$$\Phi(a, b) := \frac{1}{2} \log_2 \frac{Q_1(a, b) + \nu(a, b)}{\nu(a, b)}, \quad \nu(a, b) = D - h(a, b),$$

are compact, and the infimum in (26) is attained at some (a, b) with $h(a, b) < D$: an interior stationary point of Φ .

The stationarity system is linear in (a, b) at fixed g . At a stationary point write $g := (Q_1 + \nu)/\nu$ and $\mu_c := (a\rho + b)/s$; note $g > 1$ because $Q_1 > 0$. The partial derivatives of the three quadratics are

$$\begin{aligned} \partial_a Q_0 &= 2(a + b\rho), & \partial_b Q_0 &= 2(b + a\rho), & \partial_a h &= \partial_a Q_0 - 2, & \partial_b h &= \partial_b Q_0 - 2\rho, \\ \partial_a Q_1 &= \partial_a Q_0 - 2\rho\mu_c, & \partial_b Q_1 &= \partial_b Q_0 - 2\mu_c, \end{aligned}$$

hence the four gradient identities

$$\partial_a(Q_0 - h) = 2, \quad \partial_b(Q_0 - h) = 2\rho, \quad \partial_a(Q_1 - h) = 2(1 - \rho\mu_c), \quad \partial_b(Q_1 - h) = 2(\rho - \mu_c). \quad (27)$$

(Only the last two enter here; all four enter the weighted system of Section V.) With $\nu = D - h$, so $\partial_x \nu = -\partial_x h$, stationarity of $2 \ln 2 \cdot \Phi = \ln(Q_1 + \nu) - \ln \nu$ in $x \in \{a, b\}$ reads

$$\frac{\partial_x(Q_1 - h)}{Q_1 + \nu} + \frac{\partial_x h}{\nu} = 0, \quad \text{that is,} \quad \frac{1}{g} \partial_x(Q_1 - h) + \partial_x h = 0.$$

By (27) and $\partial_a h = 2(a + b\rho - 1)$, $\partial_b h = 2(a\rho + b - \rho)$, the two equations are

$$a + b\rho = 1 - \frac{1 - \rho\mu_c}{g}, \quad a\rho + b = \rho - \frac{\rho - \mu_c}{g}. \quad (28)$$

Substituting $\mu_c = (a\rho + b)/s$ makes (28) a 2×2 linear system in (a, b) whose coefficients depend on g alone, and it solves in closed form. The second equation reads $gs\mu_c - \mu_c = \rho(g - 1)$, so with $k := gs - 1 > 0$,

$$\mu_c = \frac{\rho(g - 1)}{k}. \quad (29)$$

Subtracting ρ times the first equation of (28) from the second gives $(a\rho + b) - \rho(a + b\rho) = b(1 - \rho^2)$ on the left and $\mu_c(1 - \rho^2)/g$ on the right, so $b = \mu_c/g$; the first equation then yields $a = 1 - (1 - \rho\mu_c)/g - b\rho = 1 - 1/g$. The stationary channel coefficients at level g are therefore

$$1 - a = \frac{1}{g}, \quad b = \frac{\mu_c}{g} = \frac{(g - 1)\rho}{gk}. \quad (30)$$

Reduction to the quadratic. At the values (30), $(1 - a)Y - bV = (Y - \mu_c V)/g$, so

$$h = \frac{1 - 2\rho\mu_c + \mu_c^2}{g^2}, \quad (31)$$

and, writing $aY + bV = Y - (Y - \mu_c V)/g$ and using the definition $a\rho + b = s\mu_c$,

$$Q_0 = 1 - \frac{2(1 - \rho\mu_c)}{g} + h, \quad Q_1 = Q_0 - \frac{(a\rho + b)^2}{s} = 1 - \frac{2(1 - \rho\mu_c)}{g} + h - s\mu_c^2.$$

The definition of g and the active constraint combine to $(g-1)(D-h) = Q_1$. Substituting the display for Q_1 and collecting the h terms ($h + (g-1)h = gh = (1 - 2\rho\mu_c + \mu_c^2)/g$ by (31)), the terms in $\rho\mu_c$ cancel and

$$1 + \frac{\mu_c^2 - 1}{g} - s\mu_c^2 = (g-1)D. \quad (32)$$

Finally substitute (29). Multiplying (32) by gk^2 and using $k^2\mu_c^2 = \rho^2(g-1)^2$,

$$gk^2 + \rho^2(g-1)^2 - k^2 - gs\rho^2(g-1)^2 = g(g-1)Dk^2.$$

Since $1 - gs = -k$, the second and fourth terms on the left combine to $-\rho^2(g-1)^2k$, so the left side is $(g-1)k[k - \rho^2(g-1)]$, and dividing by $(g-1)k > 0$,

$$k - \rho^2(g-1) = gDk \iff Dsg^2 - (D+s-\rho^2)g + (1-\rho^2) = 0,$$

that is, $P(g) = 0$. Every interior stationary point of Φ therefore sits at a root of P exceeding 1.

Root bracketing and uniqueness. $P(1) = Ds - (D+s-\rho^2) + (1-\rho^2) = (D-1)(s-1) = (D-1)\tau^2 < 0$ for $D < 1$, $\tau^2 > 0$. The leading coefficient Ds is positive and $P(0) = 1 - \rho^2 > 0$, so P has two real roots, exactly one exceeding 1 (the other lies in $(0, 1)$). Hence every stationary point has $g = g^*$, and by (30) the stationary (a, b) , and with it $\nu = Q_1/(g^* - 1)$, is unique. The attained minimum of (26) is $\frac{1}{2} \log_2 g^*$.

Achievability. The channel (21) has exactly the moments (30) with residual variance $\nu = Q_1/(g^* - 1)$. Its distortion is $h + \nu$, and $(g^* - 1)\nu = Q_1$ together with $P(g^*) = 0$ is, by the chain of equivalences above run backward, the statement $h + \nu = D$. The channel is jointly Gaussian with (T, S) , so the equality clause of Lemma 13 gives

$$I(T; \hat{Y} | S) = \frac{1}{2} \log_2 \frac{Q_1 + \nu}{\nu} = \frac{1}{2} \log_2 g^*.$$

Hence $L(D) = \frac{1}{2} \log_2 g^*$. □

Corollary 16. (i) At $\rho = 0$, $P(g) = (Dg-1)(sg-1)$ and $g^* = 1/D$: $L(D) = \frac{1}{2} \log_2(1/D)$ is the classical rate-distortion function, attained with $b = 0$ by the classical reverse test channel. (ii) As $\tau^2 \rightarrow 0$, $L(D) \rightarrow \frac{1}{2} \log_2^+((1-\rho^2)/D)$, Gray's conditional rate-distortion function of the pair (Y, V) [4], matching Proposition 12. (iii) As $\rho^2 \rightarrow 1$, $L(D) \rightarrow \frac{1}{2} \log_2[(D+\tau^2)/(D(1+\tau^2))]$, Steinberg's scalar Gaussian formula at the induced correlation $\rho_{Y,S}^2 = 1/s$ [11].

Proof. (i) $(Dg-1)(sg-1) = Dsg^2 - (D+s)g + 1 = P(g)|_{\rho=0}$. The roots are $1/D > 1$ and $1/s < 1$, so $g^* = 1/D$. At this root (21) gives $b = 0$, $a = 1 - D$, $Q_1 = (1 - D)^2$, and $\text{Var } N = (1 - D)^2/(1/D - 1) = D(1 - D)$: the classical reverse channel.

(ii) At $s = 1$, $P(g) = Dg^2 - (D+1-\rho^2)g + (1-\rho^2) = (g-1)(Dg - (1-\rho^2))$, with discriminant $(D - (1-\rho^2))^2$, so the larger root at $s = 1$ is exactly $\max\{1, (1-\rho^2)/D\}$; the roots of P are continuous in s , and the limit follows. For $D < 1 - \rho^2$ the value is $\frac{1}{2} \log_2(\text{Var}(Y | V)/D)$ with $\text{Var}(Y | V) = 1 - \rho^2$, Gray's conditional function; for $D \geq 1 - \rho^2$ it is 0.

(iii) At $\rho^2 = 1$ the constant term of P vanishes: $P(g) = g(Dsg - (D + \tau^2))$, with roots 0 and $(D + \tau^2)/(Ds)$; the latter exceeds 1 because $(D + \tau^2) - Ds = \tau^2(1 - D) > 0$. By root continuity $g^* \rightarrow (D + \tau^2)/(Ds)$ as $\rho^2 \rightarrow 1$. Steinberg's scalar function at unit source variance and side-information correlation $\rho_{Y,S}^2$ is $\frac{1}{2} \log_2[(1 - \rho_{Y,S}^2 + \rho_{Y,S}^2 D)/D]$ [11]; at $\rho_{Y,S}^2 = \rho^2/s \rightarrow 1/s$ this is $\frac{1}{2} \log_2[(\tau^2 + D)/(sD)]$, the same value. □

Corollary 17. Under the standing model, for every $\rho \neq 0$ and $0 < D < 1$, $L(D) < \frac{1}{2} \log_2 g_m$ with $g_m = (s - \rho^2 + \rho^2 D)/(sD)$, the value of Steinberg's scalar formula [11] on the reduced pair (Y, S) ; so a channel that depends on X through Y alone is strictly suboptimal.

Proof. The marginalized class attains exactly $\frac{1}{2} \log_2 g_m$, the scalar-corner value of the reduction preceding Proposition 12 at $\rho_{Y,S}^2 = \rho^2/s$. Substituting into (20) gives $P(g_m) = \rho^2(1 - D)(1 - \rho^2)/s > 0$ and $g_m - 1 = (1 - D)(s - \rho^2)/(Ds) > 0$. Since $P(1) = (D - 1)\tau^2 < 0$, the two roots of P bracket 1; a point above 1 at which $P > 0$ lies above the larger root, so $g_m > g^*$. □

V. THE RATE-CONTENT REGION AND MISALIGNMENT

The content-only endpoint of Theorem 15 does not come for free in rate. This section characterizes the full tradeoff between the two coordinates of (11) for the Gaussian pair source. Throughout, $|\rho| < 1$, $\tau^2 > 0$, $0 < D < 1$, and h , Q_0 , Q_1 , μ_c are as in the proof of Theorem 15. In the frontier system below, $\alpha \in [0, 1]$ weights rate against conditional content, and $\gamma_0 = \nu/(Q_0 + \nu)$ and $\gamma_1 = \nu/(Q_1 + \nu)$ are the two water levels the stationarity system balances: the frontier point at weight α is $(\frac{1}{2} \log_2(1/\gamma_0), \frac{1}{2} \log_2(1/\gamma_1))$. The name is used in a narrow sense: γ_0 and γ_1 are called water levels because they play the inverse signal-plus-noise roles in the two mutual-information coordinates; unlike classical parallel-channel water-filling, in which independent subchannels receive separate allocations, the two levels here are coupled through one reproduction and one distortion constraint.

We now prove part (b) of Theorem 15, the region and its frontier.

Proof of part (b). Convexity and scalarization. Both coordinates are convex in the test channel, by Lemma 3 in the general-alphabet form of Remark 4. The achievable set of pairs at distortion D is therefore convex, closed, and upward closed in each coordinate, so every point of its Pareto frontier minimizes one of the weighted objectives

$$J_\alpha := \alpha I(T; \hat{Y}) + (1 - \alpha) I(T; \hat{Y} | S), \quad \alpha \in [0, 1],$$

over admissible channels, by the supporting-hyperplane argument. At the interior weights each J_α -minimizer is a frontier point; at the endpoint weights $\alpha \in \{0, 1\}$ this a priori fixes only one coordinate of the minimizer, and the value of the other coordinate is pinned by the uniqueness of the minimizing moments (Proposition 19 below; the argument is run in detail in the proof of Corollary 20 later in this section).

The two determinant identities. Fix an admissible channel, centered without loss, with regression coordinates (a, b, ν) , $\nu > 0$, as in the proof of part (a); the Markov moment identity $\text{Cov}(N', S) = 0$ holds as there. Lemma 13 applies twice to the same three moments, once with the conditioner S and once with S deleted:

$$I(T; \hat{Y}) \geq \frac{1}{2} \log_2 \frac{\det \Sigma_T}{\det \Sigma_{e0}} = \frac{1}{2} \log_2 \frac{Q_0 + \nu}{\nu}, \quad I(T; \hat{Y} | S) \geq \frac{1}{2} \log_2 \frac{\det \Sigma_{T|S}}{\det \Sigma_e} = \frac{1}{2} \log_2 \frac{Q_1 + \nu}{\nu}, \quad (33)$$

where Σ_{e0} is the error covariance of the best linear estimate of T from $\hat{Y}$ alone. The conditional identity is written out in the proof of part (a). The rate identity, in full: with $\hat{c} := \mathbb{E}[T\hat{Y}] = \Sigma_T(a, b)^\top$ and $\text{Var } \hat{Y} = Q_0 + \nu$,

$$\Sigma_{e0} = \Sigma_T - \frac{\hat{c} \hat{c}^\top}{Q_0 + \nu}, \quad \det \Sigma_{e0} = \det \Sigma_T \left(1 - \frac{\hat{c}^\top \Sigma_T^{-1} \hat{c}}{Q_0 + \nu}\right) = \det \Sigma_T \frac{\nu}{Q_0 + \nu},$$

by the rank-one determinant identity $\det(\Sigma - \hat{c} \hat{c}^\top / \nu) = \det \Sigma (1 - \hat{c}^\top \Sigma^{-1} \hat{c} / \nu)$ and $\hat{c}^\top \Sigma_T^{-1} \hat{c} = (a, b) \Sigma_T (a, b)^\top = Q_0$.

The weighted moment program. Combining, J_α is bounded below by

$$\min_{(a, b, \nu)} \varphi_\alpha(a, b, \nu) := \frac{\alpha}{2} \log_2 \frac{Q_0 + \nu}{\nu} + \frac{1 - \alpha}{2} \log_2 \frac{Q_1 + \nu}{\nu} \quad \text{s.t.} \quad h(a, b) + \nu \leq D, \quad \nu > 0. \quad (34)$$

As in the proof of part (a): φ_α is strictly decreasing in ν at fixed $(a, b) \neq (0, 0)$, so the distortion constraint is active at the optimum; on the feasible set $Q_0 \geq Q_1 \geq q_0 > 0$, so $\varphi_\alpha \rightarrow \infty$ as $\nu \rightarrow 0$ and the minimum is attained at an interior stationary point of $\varphi_\alpha(a, b, D - h(a, b))$.

The gradient identities reduce the first-order system linearly. At a stationary point set $\gamma_0 = \nu/(Q_0 + \nu)$ and $\gamma_1 = \nu/(Q_1 + \nu)$, both in $(0, 1)$. For $x \in \{a, b\}$, using $\partial_x \nu = -\partial_x h$,

$$\alpha \frac{\partial_x(Q_0 - h)}{Q_0 + \nu} + (1 - \alpha) \frac{\partial_x(Q_1 - h)}{Q_1 + \nu} + \frac{\partial_x h}{\nu} = 0;$$

multiplying by $\nu/2$ and inserting the four gradient identities (27):

$$x = a : \quad \alpha \gamma_0 + (1 - \alpha) \gamma_1 (1 - \rho \mu_c) + (a + b \rho - 1) = 0, \quad (35)$$

$$x = b : \quad \alpha \gamma_0 \rho + (1 - \alpha) \gamma_1 (\rho - \mu_c) + (a \rho + b - \rho) = 0. \quad (36)$$

Subtract ρ times (35) from (36): the γ_0 terms cancel, the γ_1 terms give $(1 - \alpha) \gamma_1 [(\rho - \mu_c) - \rho(1 - \rho \mu_c)] = -(1 - \alpha) \gamma_1 \mu_c (1 - \rho^2)$, and the moment terms give $(a \rho + b) - \rho(a + b \rho) = b(1 - \rho^2)$, so

$$b(1 - \rho^2) = (1 - \alpha) \gamma_1 \mu_c (1 - \rho^2) \implies b = (1 - \alpha) \gamma_1 \mu_c,$$

dividing by $1 - \rho^2 > 0$, which is in force since $|\rho| < 1$ throughout this section. Equation (35) then reads $a + b\rho = 1 - \alpha\gamma_0 - (1 - \alpha)\gamma_1 + (1 - \alpha)\gamma_1\rho\mu_c = 1 - \alpha\gamma_0 - (1 - \alpha)\gamma_1 + b\rho$, so $a = 1 - \alpha\gamma_0 - (1 - \alpha)\gamma_1$. Finally $s\mu_c = a\rho + b = a\rho + (1 - \alpha)\gamma_1\mu_c$ rearranges to $\mu_c = a\rho/(s - (1 - \alpha)\gamma_1)$, the denominator being positive since $s > 1 > (1 - \alpha)\gamma_1$. This is (22); (23) restates the activity of the constraint and the definitions of (γ_0, γ_1) . Conversely, every solution of the system with $\nu > 0$ is a stationary point of the reduced program, and by Proposition 19 below the program is convex in moment coordinates with a unique minimizer, so the system's solution is unique and is that minimizer.

Achievability. The jointly Gaussian channel $\hat{Y} = aY + bV + N$, $N \sim \mathcal{N}(0, \nu)$, with the optimal (a, b, ν) realizes the same moments and, by the equality clause of Lemma 13, attains both bounds of (33) simultaneously: $I(T; \hat{Y}) = \frac{1}{2} \log_2(1/\gamma_0)$ and $I(T; \hat{Y} | S) = \frac{1}{2} \log_2(1/\gamma_1)$. Hence the value of (34) equals the minimum of J_α , the frontier point is as displayed, and the trace over $\alpha \in [0, 1]$ is the entire Pareto boundary.

Endpoints. At $\alpha = 1$, (22) gives $b = 0$ and $a = 1 - \gamma_0$; then $h = (1 - a)^2 = \gamma_0^2$, $Q_0 = a^2$, and $\gamma_0(Q_0 + \nu) = \nu$ forces $\gamma_0 a^2 = \nu(1 - \gamma_0) = \nu a$, so $\nu = \gamma_0 a$; combined with $\nu = D - \gamma_0^2$ this gives $\gamma_0(1 - \gamma_0) = D - \gamma_0^2$, that is, $\gamma_0 = D$, $a = 1 - D$, $\nu = D(1 - D)$: the classical reverse channel. At $\alpha = 0$ the γ_0 terms drop from (35)–(36), which become (28) with $g = 1/\gamma_1$; part (a) identifies $\gamma_1 = 1/g^*$ and the channel (21). $\square$

Uniqueness at every weight is a convexity statement in moment coordinates. It is cleanest, and no harder, in the whitened frame that Section VI also uses; we set it up once. Since $\Sigma_T \succ 0$ and $0 \prec \Sigma_{T|S} \preceq \Sigma_T$, there is a nonsingular W with

$$W\Sigma_T W^\top = I, \quad W\Sigma_{T|S} W^\top = \Lambda = \text{diag}(\lambda_1, \dots, \lambda_p), \quad 0 \prec \Lambda \preceq I \quad (37)$$

(factor $\Sigma_T = CC^\top$, diagonalize $C^{-1}\Sigma_{T|S}C^{-\top} = O\Lambda O^\top$, and set $W = O^\top C^{-1}$). Write $Z = WT$, so $\text{Cov } Z = I$; the described variable becomes $Y = y_0^\top Z$ with $y_0 = W^{-\top} e_Y$ and $|y_0|^2 = e_Y^\top \Sigma_T e_Y = 1$. A channel with regression vector $(a, b)^\top$ on T has regression vector $c = W^{-\top}(a, b)^\top$ on Z , and

$$Q_0 = |c|^2, \quad Q_1 = c^\top \Lambda c, \quad h = |y_0 - c|^2, \quad \mathbb{E}(Y - \hat{Y})^2 = |y_0 - c|^2 + \nu.$$

Here $p = 2$; Section VI uses $p = 1 + r$.

Lemma 18. *Let $\mathcal{Z} \subseteq \mathbb{R}^q$ be convex, $u : \mathcal{Z} \rightarrow \mathbb{R}^p$ affine, and $\sigma : \mathcal{Z} \rightarrow (0, \infty)$ concave. Then*

$$F(z) := \frac{u(z)u(z)^\top}{\sigma(z)}$$

is matrix-convex: $F(z_t) \preceq (1 - t)F(z_0) + tF(z_1)$ for all $z_0, z_1 \in \mathcal{Z}$, $t \in [0, 1]$, $z_t = (1 - t)z_0 + tz_1$.

Proof. For $\sigma > 0$ the Schur-complement criterion [36, Sec. A.5.5] states

$$\begin{bmatrix} G & u \\ u^\top & \sigma \end{bmatrix} \succeq 0 \iff G \succeq \frac{uu^\top}{\sigma},$$

so $F(z)$ is the smallest matrix admitted by the criterion at z . Fix z_0, z_1, t and abbreviate $u_i = u(z_i)$, $\sigma_i = \sigma(z_i)$, $F_i = F(z_i)$. The two positive semidefinite block matrices

$$\begin{bmatrix} F_0 & u_0 \\ u_0^\top & \sigma_0 \end{bmatrix} \succeq 0, \quad \begin{bmatrix} F_1 & u_1 \\ u_1^\top & \sigma_1 \end{bmatrix} \succeq 0$$

average, with weights $(1 - t, t)$, to

$$\begin{bmatrix} (1 - t)F_0 + tF_1 & u(z_t) \\ u(z_t)^\top & (1 - t)\sigma_0 + t\sigma_1 \end{bmatrix} \succeq 0,$$

using affinity of u in the off-diagonal blocks. Concavity gives $\sigma(z_t) \geq (1 - t)\sigma_0 + t\sigma_1$, and adding the positive semidefinite matrix $\text{diag}(0, \sigma(z_t) - (1 - t)\sigma_0 - t\sigma_1)$ raises the corner:

$$\begin{bmatrix} (1 - t)F_0 + tF_1 & u(z_t) \\ u(z_t)^\top & \sigma(z_t) \end{bmatrix} \succeq 0.$$

By the criterion again, $(1 - t)F_0 + tF_1 \succeq u(z_t)u(z_t)^\top / \sigma(z_t) = F(z_t)$. $\square$

Proposition 19. *In the whitened frame (37), with $0 \prec \Lambda \preceq I$, $|y_0| = 1$, $D \in (0, 1)$, and any $\alpha \in [0, 1]$, the weighted program (34) (with $Q_0 = |c|^2$, $Q_1 = c^\top \Lambda c$, $h = |y_0 - c|^2$, minimized over (c, ν)) has a unique minimizer, and every stationary point with $\nu > 0$ is that minimizer. Since the change of variables $c = W^{-\top}(a, b)^\top$ is a linear bijection, the same holds for the program in the coordinates of Theorem 15.*

Proof. Work in the moment coordinates $z = (c, v)$, $v = \text{Var } \hat{Y} = |c|^2 + \nu$, on the domain

$$\mathcal{D} = \{(c, v) : v > |c|^2, 1 - 2y_0^\top c + v \leq D\},$$

which is convex: $v > |c|^2$ is the strict epigraph of a convex function, and the distortion constraint $h + \nu = 1 - 2y_0^\top c + v \leq D$ is linear in z . On $\mathcal{D}$ define $\sigma := v - c^\top(I - \Lambda)c = Q_1 + \nu > 0$ and

$$B_0 = \frac{1}{2} \log_2 \frac{Q_0 + \nu}{\nu} = -\frac{1}{2} \log_2 \det \left(I - \frac{cc^\top}{\nu} \right),$$

$$B_1 = \frac{1}{2} \log_2 \frac{Q_1 + \nu}{\nu} = \frac{1}{2} \log_2 \det \Lambda - \frac{1}{2} \log_2 \det \left(\Lambda - \frac{(\Lambda c)(\Lambda c)^\top}{\sigma} \right);$$

the first equality holds because $\det(I - cc^\top/\nu) = 1 - |c|^2/\nu = \nu/\nu$, the second because $\det(\Lambda - (\Lambda c)(\Lambda c)^\top/\sigma) = \det \Lambda (1 - c^\top \Lambda c/\sigma) = \det \Lambda \cdot \nu/\sigma$.

Convexity. Apply Lemma 18 twice. For B_0 : $z \mapsto c$ is affine and $z \mapsto v$ is affine, hence concave and positive on $\mathcal{D}$, so $G_0(z) = cc^\top/\nu$ is matrix-convex. Moreover $I - G_0 \succ 0$ on $\mathcal{D}$, hence invertible: by the Schur criterion, $I \succ cc^\top/\nu$ is equivalent to $v - |c|^2 > 0$, and $v - |c|^2 = \nu > 0$ on $\mathcal{D}$ by construction. For B_1 : $z \mapsto \Lambda c$ is affine and $z \mapsto \sigma = v - c^\top(I - \Lambda)c = Q_1 + \nu$ is concave (affine minus a convex quadratic, $I - \Lambda \succeq 0$) and positive on $\mathcal{D}$ ($Q_1 \geq 0$, $\nu > 0$), so $G_1(z) = (\Lambda c)(\Lambda c)^\top/\sigma$ is matrix-convex, $\Lambda - G_1$ is matrix-concave, and $\Lambda - G_1 \succ 0$ because $c^\top \Lambda c = Q_1 < Q_1 + \nu = \sigma$. The map $\Sigma \mapsto -\log_2 \det \Sigma$ on positive definite matrices is convex and decreasing in the Löwner order; for the latter, if $0 \prec A \preceq B$ then $B^{-1/2}AB^{-1/2} \preceq I$ has eigenvalues in $(0, 1]$, so $\det A/\det B = \det(B^{-1/2}AB^{-1/2}) \leq 1$, that is, $\det A \leq \det B$. Composing a convex antitone function with a matrix-concave map preserves convexity: for matrix-concave $F \succ 0$,

$$-\log \det F(z_t) \leq -\log \det((1-t)F(z_0) + tF(z_1)) \leq (1-t)(-\log \det F(z_0)) + t(-\log \det F(z_1)).$$

Hence B_0 , B_1 , and every $B_\alpha := \alpha B_0 + (1-\alpha)B_1$ are convex on $\mathcal{D}$.

Reduction to the active slice. On $\mathcal{D}$, feasibility forces $c \neq 0$: from $v > |c|^2 \geq 0$ and $1 - 2y_0^\top c + v \leq D$,

$$2y_0^\top c \geq 1 - D + v > 1 - D > 0. \quad (38)$$

At fixed $c \neq 0$ both B_0 and B_1 strictly decrease in v ($\partial_v[\ln v - \ln \nu] = v^{-1} - \nu^{-1} < 0$ and $\partial_v[\ln \sigma - \ln \nu] = \sigma^{-1} - \nu^{-1} < 0$, since $Q_0 \geq Q_1 > 0$), so every minimizer of B_α lies on the active slice $v = D - 1 + 2y_0^\top c$, an affine section of $\mathcal{D}$. On the slice B_α is convex, so its stationary points are exactly its global minimizers and the minimizer set is convex; coercivity as in the proof of Theorem 15(b) gives attainment.

Strictness. Suppose $z_0 \neq z_1$ are both minimizers on the slice, and let z_t run over the segment, along which v and c are affine in t . Then B_α is affine on the segment, and since B_0 , B_1 are convex and their weighted sum is affine, each B_i carrying nonzero weight is affine on the segment. We use twice the following fact: for an affine positive definite matrix segment $\Sigma_t = (1-t)\Sigma_0 + t\Sigma_1$, the function $t \mapsto -\log \det \Sigma_t$ has second derivative $\text{tr}[(\Sigma_t^{-1}\dot{\Sigma})^2] \geq 0$, which vanishes only if $\dot{\Sigma} = 0$, because $\Sigma_t^{-1/2}\dot{\Sigma}\Sigma_t^{-1/2}$ is symmetric and a symmetric matrix whose square has zero trace is zero. Moreover, if a matrix-concave F has $-\log \det F$ affine on the segment, then squeezing $-\log \det F(z_t) \leq -\log \det((1-t)F(z_0) + tF(z_1)) \leq$ the affine interpolant forces both inequalities to be equalities; the second forces $F(z_0) = F(z_1) =: \bar{F}$ by the fact just stated, and the first then forces $F(z_t) \succeq \bar{F}$ with $\det F(z_t) = \det \bar{F}$, hence $F(z_t) = \bar{F}$ for all t (if $A \succeq B \succ 0$ and $\det A = \det B$ then $A = B$).

Case $\alpha > 0$, where strictness comes from the forced constancy of the rank-one ratio G_0 along the segment: B_0 is affine, so $G_0(z_t) = c(t)c(t)^\top/v(t)$ is constant, a rank-one matrix $\bar{G}_0 \neq 0$ by (38). Then $c(t) = \theta(t)e$ for a fixed unit vector e , with θ affine (a coordinate of the affine c), and $\theta(t)^2 = v(t)(e^\top \bar{G}_0 e)$ with v affine. An affine function whose square is affine is constant, so θ , hence c , hence $v = D - 1 + 2y_0^\top c$, is constant: $z_0 = z_1$, a contradiction.

Case $\alpha < 1$, where strictness comes from the constancy of the rank-one ratio G_1 played against the concavity of σ : B_1 is affine, so $G_1(z_t) = (\Lambda c)(\Lambda c)^\top/\sigma$ is constant and nonzero. Then $\Lambda c(t) = \theta(t)e$ with θ affine, and $\theta(t)^2 = \sigma(t)(e^\top \bar{G}_1 e)$. Along the segment $\sigma(t) = v(t) - c(t)^\top(I - \Lambda)c(t)$ is a concave quadratic in t , while θ^2 is

a convex quadratic; equality up to the positive constant $e^\top \bar{G}_1 e$ forces both to be affine, so θ is an affine function with affine square, hence constant. Then Λc is constant, c is constant (Λ nonsingular), and $z_0 = z_1$ as before, a contradiction.

Every $\alpha \in [0, 1]$ falls in at least one case, so the minimizer is unique.

We record one further fact for the clean-context boundary. At $\alpha = 1$ the argument is Λ -free: the objective is B_0 alone, whose convexity (Lemma 18 with $u(z) = c$ and $\sigma = v$), strict decrease in v at $c \neq 0$, coercivity through $Q_0 \geq q_0 > 0$, and strictness case use only c, v , and the domain $\mathcal{D}$, never Λ . The uniqueness of the rate minimizer, equivalently of the moment triple $(a, b, \nu) = (1 - D, 0, D(1 - D))$, therefore holds for an arbitrary conditioner, including the boundary $\tau^2 = 0$, where $\Sigma_{T|S}$ is singular and the whitening (37) of the conditional covariance is unavailable. $\square$

Corollary 20. *Within the standing model of Section III ($\tau^2 > 0, |\rho| < 1, 0 < D < 1$), the frontier is nondegenerate if and only if $\rho^2 \in (0, 1)$. When $\rho^2 \in (0, 1)$: any channel attaining $L(D)$ has rate strictly above $R_{\min} = \frac{1}{2} \log_2(1/D)$, and any channel attaining $R_{\min}$ has conditional content strictly above $L(D)$. In particular, writing $L_R^*(D)$ for the conditional content of the rate-optimal channel and $R_L^*(D)$ for the rate of the content-optimal channel,*

$$L_R^*(D) = \frac{1}{2} \log_2 \frac{(1-D)(1-\rho^2/s) + D}{D} > L(D),$$

the content of the classical reverse channel, and $R_L^(D) = \frac{1}{2} \log_2[(Q_0 + \nu)/\nu]$ evaluated at the channel of Theorem 15 strictly exceeds $R_{\min}$. When $\rho = 0$ the classical reverse channel attains both minima and the frontier is a single corner. The frontier also collapses in the limits $\rho^2 \rightarrow 1$ (where both endpoint excesses vanish; see the proof) and $\tau^2 \rightarrow \infty$ (where $L \rightarrow R$ coordinatewise); positivity of τ^2 , by contrast, is a modeling assumption and not necessary for the tradeoff (Proposition 21).*

Proof. By part (a) of Theorem 15, the moment program for the conditional coordinate alone has the unique minimizer (a^*, b^*, ν^*) of (30), and $b^* = (g^* - 1)\rho/(g^*k) \neq 0$ since $\rho \neq 0, g^* > 1, k > 0$. By the $\alpha = 1$ endpoint computation in the proof of part (b) (uniqueness by Proposition 19), the moment program for the rate coordinate alone has the unique minimizer $(1 - D, 0, D(1 - D))$, with $b = 0$. The two minimizers are distinct.

Let a channel attain $L(D)$. Writing B_0, B_1 for the two bounds of (33) as functions of the channel's moments, $L(D) = I(T; \hat{Y} | S) \geq B_1$ (its moments) $\geq \min B_1 = L(D)$ forces equality throughout, so its moments minimize B_1 and equal (a^*, b^*, ν^*) . Its rate then satisfies $R \geq B_0(a^*, b^*, \nu^*) > \min B_0 = R_{\min}$: strict, because B_0 's minimizer is unique and differs from (a^*, b^*, ν^*) . Symmetrically, any rate-optimal channel carries the moments $(1 - D, 0, D(1 - D))$ and its content is at least B_1 there, which strictly exceeds $L(D)$; evaluating, $b = 0$ gives $Q_1 = (1 - D)^2(1 - \rho^2/s)$ and $\nu = D(1 - D)$, so $(Q_1 + \nu)/\nu = [(1 - D)(1 - \rho^2/s) + D]/D$, the displayed $L_R^*(D)$.

At $\rho = 0$, $b^* = 0$ and $g^* = 1/D$ (Corollary 16(i)) give $(a^*, \nu^*) = (1 - D, D(1 - D))$: the two minimizers coincide and the frontier is the corner $R_{\min} = L(D) = \frac{1}{2} \log_2(1/D)$. As $\rho^2 \rightarrow 1$ both endpoint excesses vanish, so the frontier collapses to a corner in the limit: by Corollary 16(iii), $L(D) \rightarrow \frac{1}{2} \log_2[(D + \tau^2)/(Ds)]$, and the displayed $L_R^*(D)$ has the same limit, since $(1 - D)(1 - \rho^2/s) + D \rightarrow (1 - D)\tau^2/s + D = (D + \tau^2)/s$; for the rate side, the channel (21) has, as $\rho \rightarrow 1$, $g^* \rightarrow (D + \tau^2)/(Ds)$ and

$$(a, b, \nu) \rightarrow \left(\frac{\tau^2(1 - D)}{D + \tau^2}, \frac{D(1 - D)}{D + \tau^2}, D(1 - D) \right),$$

whence $a + b \rightarrow 1 - D$, $Q_0 \rightarrow (1 - D)^2$, and $R_L^*(D) = \frac{1}{2} \log_2[(Q_0 + \nu)/\nu] \rightarrow \frac{1}{2} \log_2(1/D) = R_{\min}$; for $\rho \rightarrow -1$ only the sign of b flips, leaving Q_0 and R_L^* unchanged. As $\tau^2 \rightarrow \infty$, $\Sigma_{T|S} \rightarrow \Sigma_T$ makes the two coordinates, and so the two endpoints, coincide in the limit. $\square$

Fig. 2 traces the frontier at the numerical instance of Corollary 20. The displayed excesses are representative of moderate parameters, not extremal. Over the box $\rho^2 \in [0.05, 0.95]$, $\tau^2 \in [\frac{1}{4}, 4]$, $D \in [0.05, 0.95]$, the endpoint excesses reach at most $R_L^*(D) - R_{\min} = 0.1138$ bits, at $(\rho^2, \tau^2, D) = (0.5, 0.25, 0.5)$, and $L_R^*(D) - L(D) = 0.0770$ bits, at $(0.615, 0.25, 0.385)$. Toward the clean-context boundary the rate excess is unbounded: at $D = 1 - \rho^2$ the content-optimal channel pins the reproduction to the context as $\tau^2 \rightarrow 0$ and its rate diverges, reaching, for instance, $R_L^*(D) - R_{\min} = 1.537$ bits at $(0.5, 10^{-3}, 0.5)$; the content excess instead saturates there at $\frac{1}{2} \log_2(1 + \rho^2)$, the clean-boundary value of Proposition 21 evaluated at $D = 1 - \rho^2$. All quoted values are computed by the verification script.

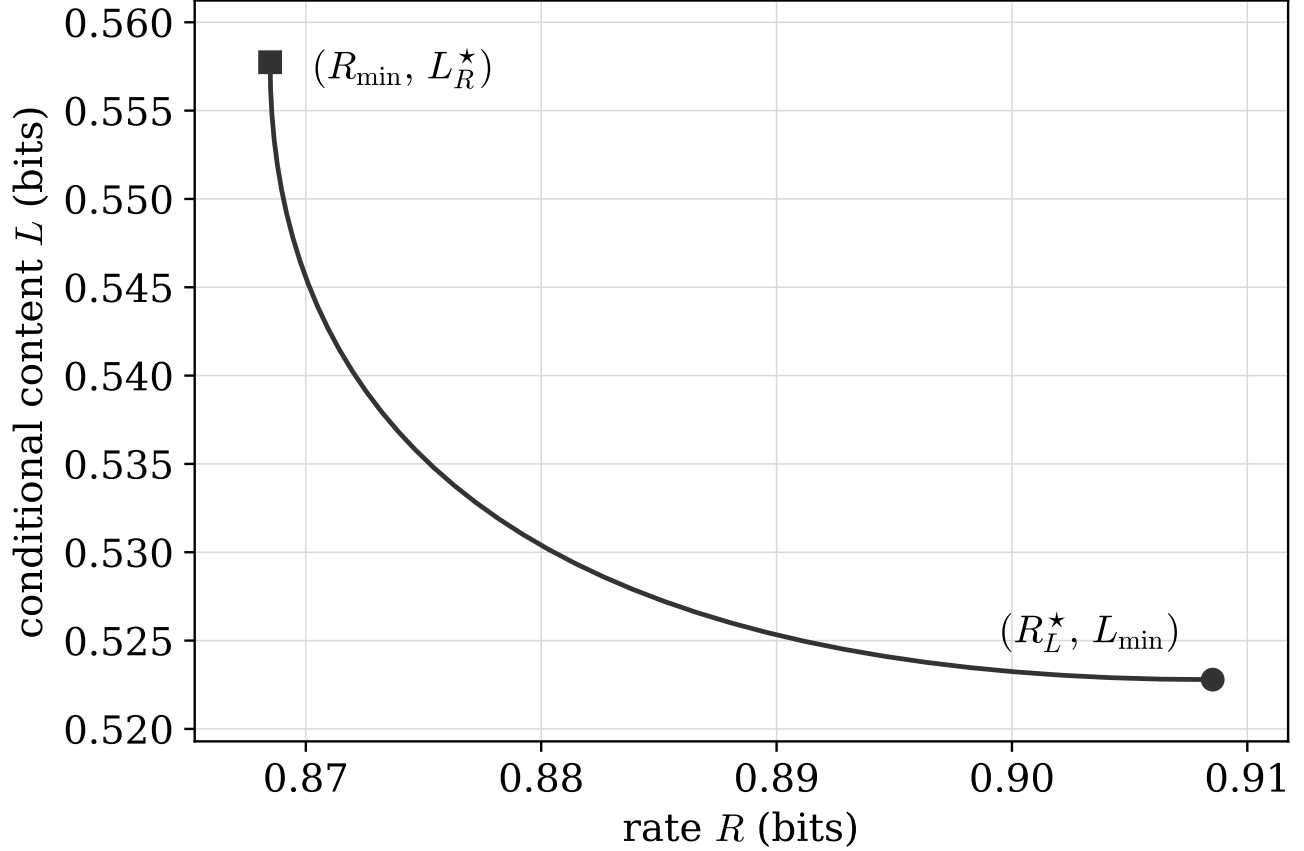


Fig. 2. The Pareto frontier of the rate-content region at $(\rho^2, \tau^2, D) = (0.75, 0.5, 0.3)$, traced by the two-water-level system of Theorem 15(b). The tradeoff is strict: the rate-optimal channel pays $L_R^*(D) - L(D) = 0.0349$ bits of excess conditional content, and the content-optimal channel pays $R_L^*(D) - R_{\min} = 0.0400$ bits of excess rate.

Fig. 3 repeats the construction at $\rho^2 \in \{0.2, 0.5, 0.8\}$ with $\tau^2 = 0.5$ and $D = 0.3$; the frontier is a nondegenerate curve at each, as Corollary 20 states.

Proposition 21. *The strict tradeoff extends to the boundary $\tau^2 = 0$ excluded by the standing model. Let $\tau^2 = 0$, so $S = V$, and let $\rho^2 \in (0, 1)$, $0 < D < 1$. The minimal conditional content is*

$$L_{\min}(D) = \frac{1}{2} \log_2^+ \frac{1 - \rho^2}{D}, \quad (39)$$

while every channel attaining the minimal rate $R_{\min} = \frac{1}{2} \log_2(1/D)$, in particular the classical reverse channel $\hat{Y}_R = (1 - D)Y + N$ with $N \sim \mathcal{N}(0, D(1 - D))$ independent of X , has conditional content exactly

$$L_R^*(D) = \frac{1}{2} \log_2 \frac{(1 - D)(1 - \rho^2) + D}{D} = \frac{1}{2} \log_2 \frac{1 - \rho^2 + D\rho^2}{D} > L_{\min}(D), \quad (40)$$

strictly, on all of $0 < D < 1$. In particular no channel attains $R_{\min}$ and $L_{\min}$ simultaneously: a clean context trades rate against content just as a noisy one does, for $D < 1 - \rho^2$ with both quantities positive, and for $D \geq 1 - \rho^2$ with $L_{\min} = 0$ but $L_R^* > 0$.

Proof. The value (39). Write $Y = \rho V + Y'$ with $Y' \perp V$, $\text{Var } Y' = 1 - \rho^2$. Converse: for any admissible channel, $I(X; \hat{Y} | V) \geq I(Y; \hat{Y} | V) \geq R_{Y|V}(D) = \frac{1}{2} \log_2^+(\text{Var}(Y | V)/D)$, Gray's conditional function [4], exactly

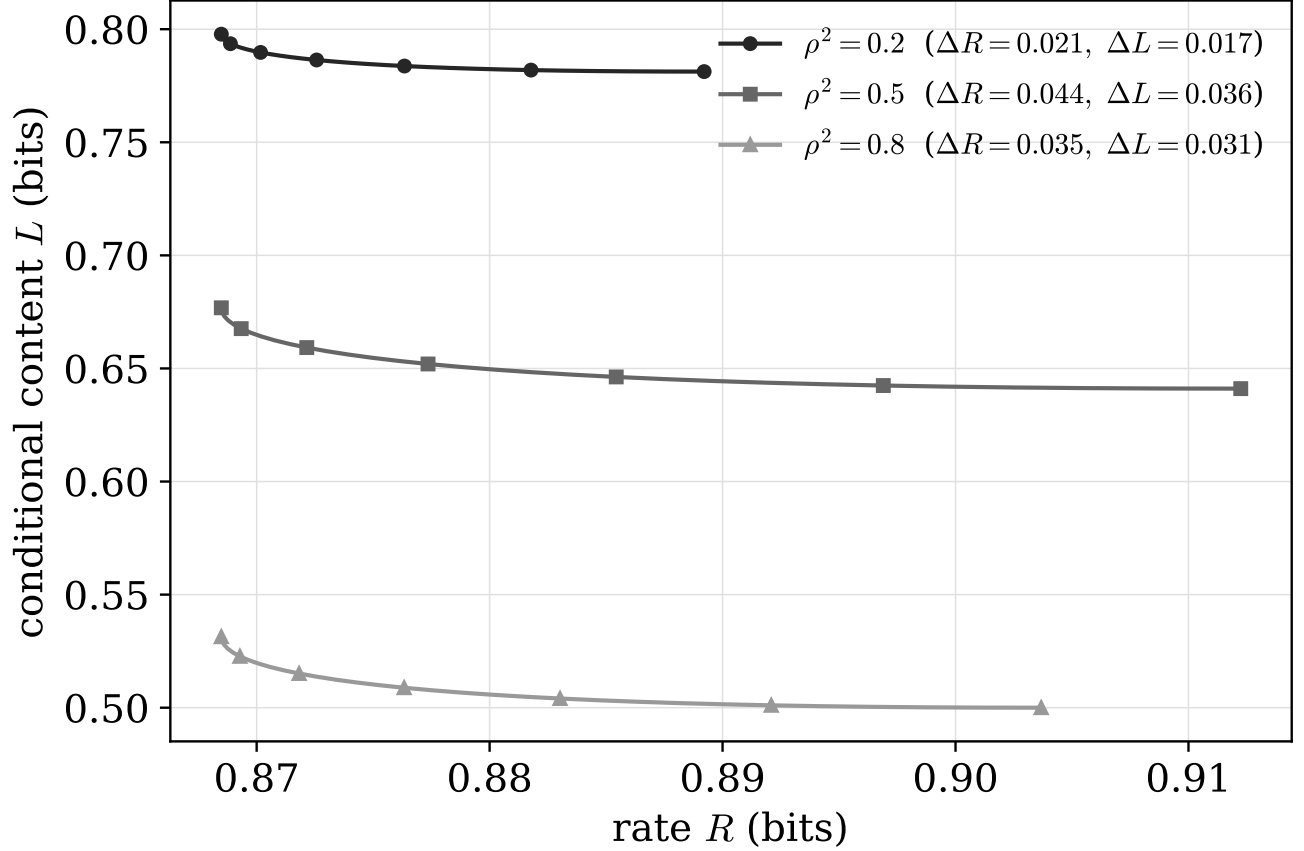


Fig. 3. The frontier at three correlations, fixed $(\tau^2, D) = (0.5, 0.3)$. The endpoint gap between the rate-optimal and content-optimal channels widens from $\rho^2 = 0.2$ ($\Delta R = 0.021, \Delta L = 0.017$ bits) to $\rho^2 = 0.5$ ($\Delta R = 0.044, \Delta L = 0.036$) and eases at $\rho^2 = 0.8$ ($\Delta R = 0.035, \Delta L = 0.031$): the tradeoff vanishes at $\rho^2 = 0$, is widest at intermediate misalignment, and closes again toward the $\rho^2 \rightarrow 1$ boundary. Marker shape distinguishes the curves; the gaps are printed by the plotting script.

as in Proposition 12(ii). Achievability for $D \leq 1 - \rho^2$: with $\theta = D/(1 - \rho^2)$, take $\hat{Y} = \rho V + (1 - \theta)Y' + N$, $N \sim \mathcal{N}(0, \theta(1 - \theta)(1 - \rho^2))$: the distortion is $\theta^2(1 - \rho^2) + \theta(1 - \theta)(1 - \rho^2) = D$, and

$$I(T; \hat{Y} | V) = \frac{1}{2} \log_2 \frac{\text{Var}(\hat{Y} | V)}{\text{Var}(\hat{Y} | T)} = \frac{1}{2} \log_2 \frac{(1 - \theta)^2(1 - \rho^2) + \theta(1 - \theta)(1 - \rho^2)}{\theta(1 - \theta)(1 - \rho^2)} = \frac{1}{2} \log_2 \frac{1 - \rho^2}{D}.$$

For $D \geq 1 - \rho^2$, $\hat{Y} = \rho V$ has distortion $1 - \rho^2 \leq D$ and is V -measurable, so $I(T; \hat{Y} | V) = 0$.

The value (40). For the reverse channel, $\text{Var}(\hat{Y}_R | V) = (1 - D)^2 \text{Var}(Y | V) + \text{Var} N = (1 - D)^2(1 - \rho^2) + D(1 - D)$ and $\text{Var}(\hat{Y}_R | T) = \text{Var} N = D(1 - D)$, so

$$L_R^*(D) = \frac{1}{2} \log_2 \frac{(1 - D)^2(1 - \rho^2) + D(1 - D)}{D(1 - D)} = \frac{1}{2} \log_2 \frac{(1 - D)(1 - \rho^2) + D}{D},$$

and $(1 - D)(1 - \rho^2) + D = 1 - \rho^2 + D\rho^2$.

Every rate-optimal channel carries the content (40). The rate moment program never references the conditioner, and at $\alpha = 1$ the uniqueness proof of Proposition 19 is Λ -free (the closing note there), so also at $\tau^2 = 0$ its unique minimizer is $(a, b, \nu) = (1 - D, 0, D(1 - D))$. Let a channel attain $R_{\min}$ at distortion D . The chain $R_{\min} = I(T; \hat{Y}) \geq B_0(\text{its moments}) \geq \min B_0 = R_{\min}$ forces its moments to be that triple and forces equality in the max-entropy bound (18). Equality makes the linear-estimation error $E := T - J\hat{Y}$, $J = \mathbb{E}[T\hat{Y}]/\text{Var} \hat{Y}$, Gaussian and independent of $\hat{Y}$. At these moments $\text{Var} \hat{Y} = (1 - D)^2 + D(1 - D) = 1 - D$ and $\mathbb{E}[Y\hat{Y}] = 1 - D$, so the Y -row of J is 1 and $Y = \hat{Y} + E_Y$ with E_Y Gaussian independent of $\hat{Y}$ and Y standard Gaussian;

TABLE II

THE VALUES BEHIND COROLLARY 22: TWO INSTANCES WITH THE SAME REDUCED (Y, S) LAW ($\rho_{YS}^2 = \frac{1}{2}$), IN BITS. THE TWO L ROWS DIFFER AT BOTH DISTORTIONS, AND BOTH LIE BELOW THE STEINBERG ROW. EXACT FORMS FOR THE SECOND INSTANCE: $g^* = (17 + \sqrt{229})/6$ AT $D = 0.1$ AND $(21 + \sqrt{261})/18$ AT $D = 0.3$ (DISPLAYED IN THE PROOF).

	$\text{Corr}(Y, S) = \rho/\sqrt{s}$	L at $D = 0.1$	L at $D = 0.3$
Instance $(\rho^2, \tau^2) = (\frac{1}{2}, 0)$	$1/\sqrt{2}$	1.1610	0.3685
Instance $(\rho^2, \tau^2) = (\frac{3}{4}, \frac{1}{2})$	$1/\sqrt{2}$	1.2105	0.5228
Steinberg's scalar formula on the reduced pair	$1/\sqrt{2}$	1.2297	0.5577

by Cramér's decomposition theorem [8], $\hat{Y}$ is Gaussian. Independent Gaussian vectors are jointly Gaussian, so $(T, \hat{Y}) = (J\hat{Y} + E, \hat{Y})$ is jointly Gaussian with the pinned moments: the law of the reverse channel. Its conditional content is then determined by that law,

$$I(T; \hat{Y} | V) = \frac{1}{2} \log_2 \frac{\text{Var}(\hat{Y} | V)}{\text{Var}(\hat{Y} | T)} = \frac{1}{2} \log_2 \frac{(1-D) - (1-D)^2 \rho^2}{D(1-D)} = \frac{1}{2} \log_2 \frac{1 - \rho^2 + D\rho^2}{D},$$

which is (40).

Strictness. For $D < 1 - \rho^2$: $1 - \rho^2 + D\rho^2 > 1 - \rho^2$ since $D\rho^2 > 0$, so $L_R^* > L_{\min}$. For $1 - \rho^2 \leq D < 1$: $L_{\min} = 0$, while $1 - \rho^2 + D\rho^2 > D \Leftrightarrow (1 - \rho^2)(1 - D) > 0$, so $L_R^* > 0$. $\square$

Corollary 22. *Two instances whose (Y, S) joint laws are equivalent up to bijective scaling can have different minimal conditional content: the instances $(\rho^2, \tau^2) = (\frac{1}{2}, 0)$ and $(\frac{3}{4}, \frac{1}{2})$ have $\rho_{YS}^2 = \frac{1}{2}$ in both cases, yet different L at both tested distortions; Table II lists the values, both strictly below Steinberg's scalar formula evaluated on the reduced pair. So no functional of the (Y, S) joint law, in particular neither Steinberg's scalar formula evaluated on (Y, S) nor Gray's, can equal L . This is no departure from Steinberg's functional, whose source here is the full pair (Y, V) ; it shows the augmentation is essential: the encoder's access to the clean context V enters the function.*

Proof. The two instances share the (Y, S) joint law: in both, Y is standard and $\rho_{YS}^2 = \text{Cov}(Y, S)^2 / \text{Var } S = \rho^2 / s$ equals $\frac{1}{2}$, namely $(\frac{1}{2})/1$ and $(\frac{3}{4})/(\frac{3}{2})$, and rescaling S changes none of the quantities considered: a bijective scaling $S \mapsto \kappa S$, $\kappa \neq 0$, leaves the generated σ -algebra unchanged, $\sigma(\kappa S) = \sigma(S)$, so every conditional mutual information given S , and with it $L(D)$, is invariant, and the two instances may be normalized to a common (Y, S) law.

First instance, $(\rho^2, \tau^2) = (\frac{1}{2}, 0)$: the context is clean. This instance sits at the boundary $\tau^2 = 0$ excluded by the standing assumption $\tau^2 > 0$ of Section III, so its value is supplied not by Theorem 15 but directly by Proposition 12(ii), which is proved at $\tau^2 = 0$ (and which the $\tau^2 \rightarrow 0$ anchor of Corollary 16 matches in the limit). The instance gives

$$L = \frac{1}{2} \log_2 \frac{1}{2D},$$

so $L = \frac{1}{2} \log_2 5 = 1.1610$ bits at $D = 0.1$, and $L = \frac{1}{2} \log_2(5/3) = 0.3685$ bits at $D = 0.3$.

Second instance, $(\rho^2, \tau^2) = (\frac{3}{4}, \frac{1}{2})$: $s = \frac{3}{2}$, and (20) becomes, at $D = 0.1$,

$$P(g) = \frac{3}{20}g^2 - \frac{17}{20}g + \frac{1}{4}, \quad g^* = \frac{17 + \sqrt{229}}{6} = 5.3555, \quad L = \frac{1}{2} \log_2 g^* = 1.2105 \text{ bits},$$

and, at $D = 0.3$,

$$P(g) = \frac{9}{20}g^2 - \frac{21}{20}g + \frac{1}{4}, \quad g^* = \frac{21 + \sqrt{261}}{18} = 2.0642, \quad L = \frac{1}{2} \log_2 g^* = 0.5228 \text{ bits}.$$

The value of Steinberg's scalar formula on the reduced pair: an encoder observing Y alone pays the marginalized-class optimum of Section IV, the scalar-corner value $\frac{1}{2} \log_2 [(1 - \rho_{YS}^2 + \rho_{YS}^2 D)/D]$ at $\rho_{YS}^2 = \frac{1}{2}$, that is, $\frac{1}{2} \log_2 [(1 + D)/(2D)]$: $\frac{1}{2} \log_2 5.5 = 1.2297$ bits at $D = 0.1$, and $\frac{1}{2} \log_2(13/6) = 0.5577$ bits at $D = 0.3$. Both instances lie strictly below it ($1.1610, 1.2105 < 1.2297$ and $0.3685, 0.5228 < 0.5577$), as Proposition 12 requires.

Since the two instances have the same (Y, S) law after this normalization and different L , no functional of the (Y, S) joint law, in particular neither Steinberg's scalar formula evaluated on the pair nor Gray's under any reparameterization, can equal L . $\square$

The mechanism is worth stating plainly. Both instances have the same reduced correlation $\text{Corr}(Y, S) = \rho/\sqrt{s} = 1/\sqrt{2}$, realized by different splits: entirely through a clean context in the first instance ($\rho^2 = \frac{1}{2}$, $\tau^2 = 0$), through a stronger context read in noise in the second ($\rho^2 = \frac{3}{4}$, $\tau^2 = \frac{1}{2}$). The function depends on (ρ^2, τ^2) separately, not only through the ratio ρ^2/s , because the encoder's channel acts on its observation of (Y, V) , not on (Y, S) : two sources with identical (Y, S) joint laws but different internal splits of the correlation admit different optimal channels. The gap at the tabulated levels is 0.0495 bits at $D = 0.1$ and 0.1543 bits at $D = 0.3$.

Remark 23. *The closed form settles when the conditional-variance lower bound is attained. Every admissible channel obeys $L \geq I(Y; \hat{Y} | S) = h(Y | S) - h(Y | \hat{Y}, S) \geq h(Y | S) - h(Y - \hat{Y}) \geq \frac{1}{2} \log_2(\text{Var}(Y | S)/D) = \frac{1}{2} \log_2 g_f$ with $g_f = (s - \rho^2)/(Ds)$, by the Gaussian maximum-entropy bound at variance D . Within the standing model $\tau^2 > 0$ the bound, taken as its positive part where $g_f < 1$, is attained exactly when $\rho = 0$: substituting g_f into (20) gives $P(g_f) = -\rho^2\tau^2/s$, negative when $\rho \neq 0$, so that $g^* > g_f$ strictly; at $\rho = 0$ the quadratic factors as $(Dg - 1)(sg - 1)$ and $g_f = 1/D$ is its larger root. The clean-context case $\tau^2 = 0$, where the bound is Gray's function and is attained on $D \leq 1 - \rho^2$, is Proposition 21.*

VI. VECTOR CONTEXT

The frontier survives a vector conditioner. Let $S = \Gamma^T X + U \in \mathbb{R}^r$, $U \sim \mathcal{N}(0, \Sigma_U)$, $\Sigma_U \succ 0$, with X , U , and all channel randomness mutually independent. Write $V := \Gamma^T X \in \mathbb{R}^r$ and $T := (Y, V) \in \mathbb{R}^{1+r}$, and assume $\Sigma_T \succ 0$; degenerate instances reduce by dropping linearly dependent context components. Reproductions are scalar without loss of optimality by the data-processing reduction of Section III, which is generic in S .

Theorem 24. *Let $0 < D < 1$. (i) Sufficiency: it is without loss of optimality to let the channel depend on X only through T , and then $I(X; \hat{Y} | S) = I(T; \hat{Y} | S)$. (ii) Whitened form: with W , $Z = WT$, $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_{1+r})$, $0 \prec \Lambda \preceq I$, and the unit vector y_0 of (37) at $p = 1 + r$, every admissible channel with regression vector $c = \mathbb{E}[Z\hat{Y}]$ and residual variance $\nu > 0$ satisfies*

$$R \geq \frac{1}{2} \log_2 \frac{|c|^2 + \nu}{\nu}, \quad L \geq \frac{1}{2} \log_2 \frac{c^T \Lambda c + \nu}{\nu}, \quad \mathbb{E}(Y - \hat{Y})^2 = |y_0 - c|^2 + \nu, \quad (41)$$

with both bounds attained simultaneously by the jointly Gaussian channel $\hat{Y} = c^T Z + N$, $N \sim \mathcal{N}(0, \nu)$. (iii) The frontier, traced by $\alpha \in [0, 1]$: with $\gamma_0 = \nu/(|c|^2 + \nu)$ and $\gamma_1 = \nu/(c^T \Lambda c + \nu)$, the Pareto-optimal channel is the unique solution with $\nu > 0$ of the generalized two-water-level system

$$c = (1 - \alpha\gamma_0 - (1 - \alpha)\gamma_1) [(1 - (1 - \alpha)\gamma_1)I + (1 - \alpha)\gamma_1\Lambda]^{-1} y_0, \quad \nu = D - |y_0 - c|^2, \quad (42)$$

per-mode explicit given the two scalars (γ_0, γ_1) , and the frontier point is $(\frac{1}{2} \log_2(1/\gamma_0), \frac{1}{2} \log_2(1/\gamma_1))$. For $r = 1$ the program coincides with that of Theorem 15(b) under the linear bijection $c = W^{-T}(a, b)^T$, so the two systems trace the same frontier. The $\alpha = 0$ endpoint gives the vector-context function $L(D)$ as a one-dimensional root problem in γ_1 .

Proof. (i) In the resampling proof of Lemma 11, the only property of S used is that its conditional law given $(X, \hat{Y})$, and given $(X, \hat{Y}')$, is the fixed kernel $p(s | t, \hat{y}) = p_U(s - v)$, depending on x only through t . This holds verbatim for the vector $S = V + U$ with $U \sim \mathcal{N}(0, \Sigma_U)$ jointly independent of X and the channel randomness. Steps (a) through (d) of that proof are unchanged.

(ii) *Whitening.* First, $\Sigma_{T|S} = \Sigma_T - \Sigma_{TS}\Sigma_S^{-1}\Sigma_{ST} \succ 0$, where $\Sigma_S = \Sigma_V + \Sigma_U$ is the conditioner's covariance (V is not renormalized in this section): positivity of $\text{Cov}(T, S)$ holds because $\Sigma_T \succ 0$ and $\Sigma_{S|T} = \text{Cov}(\Gamma^T X | T) + \Sigma_U \succeq \Sigma_U \succ 0$, and the Schur complement of a positive definite matrix is positive definite. Also $\Sigma_{T|S} \preceq \Sigma_T$. The congruence (37) therefore exists at $p = 1 + r$ with $0 \prec \Lambda \preceq I$, and $\text{Cov} Z = I$ makes y_0 a unit vector: $Y = y_0^T Z$, $|y_0|^2 = e_Y^T \Sigma_T e_Y = 1$.

Moments and the two bounds. Fix an admissible channel, centered without loss, with $c = \mathbb{E}[Z\hat{Y}]$ and $\nu = \text{Var} \hat{Y} - |c|^2$; the case $\nu = 0$ is dispatched by Remark 14 (the reproduction is then a noiseless linear functional of T , so either every bound is trivial or the reproduction is null). The residual $N' = \hat{Y} - c^T Z$ is uncorrelated with Z , hence with T and with its subvector V ; and it is uncorrelated with U by mutual independence, hence with $S = V + U$: componentwise, $\text{Cov}(N', S) = 0$. Lemma 13 (the two determinant lower bounds, with equality for the jointly Gaussian channel matching the second moments) applies with $A = c^T W$: $A \Sigma_T A^T = c^T W \Sigma_T W^T c = |c|^2$

and $A\Sigma_{T|S}A^\top = c^\top\Lambda c$, giving the two bounds of (41). The distortion evaluation is exact because Z is white: $\mathbb{E}(Y - \hat{Y})^2 = 1 - 2y_0^\top c + (|c|^2 + \nu) = |y_0 - c|^2 + \nu$. Equality for the Gaussian channel is Lemma 13's equality clause.

(iii) *The weighted program.* As in Theorem 15(b): both coordinates are convex in the channel, so the frontier is the trace of the weighted minimizations, and the weighted objective is bounded below by the moment program

$$\min_{c, \nu} \frac{\alpha}{2} \log_2 \frac{Q_0 + \nu}{\nu} + \frac{1 - \alpha}{2} \log_2 \frac{Q_1 + \nu}{\nu}, \quad Q_0 = |c|^2, \quad Q_1 = c^\top \Lambda c, \quad |y_0 - c|^2 + \nu \leq D, \quad \nu > 0,$$

whose value the Gaussian channel of (ii) attains. The objective is strictly decreasing in ν at fixed $c \neq 0$, so the distortion constraint is active at the optimum. Feasibility forces $2y_0^\top c \geq 1 - D + (|c|^2 + \nu) > 1 - D > 0$ as in (38), so $c \neq 0$ and $Q_1 \geq \lambda_{\min}|c|^2$ is bounded away from zero on the feasible set, giving coercivity as $\nu \rightarrow 0$ and attainment at an interior stationary point of the reduced objective in c , with $\nu = D - |y_0 - c|^2$.

The first-order system. Differentiating $\alpha \ln(Q_0 + \nu) + (1 - \alpha) \ln(Q_1 + \nu) - \ln \nu$ in c with $\nu = D - |y_0 - c|^2$, exactly as in the proof of Theorem 15(b) with $\nabla Q_0 = 2c$ and $\nabla Q_1 = 2\Lambda c$, and multiplying by $\nu/2$ gives (42); the bracket is invertible, each diagonal entry being a convex combination of 1 and $\lambda_j > 0$. Uniqueness of the minimizer, hence of the system's solution with $\nu > 0$, at every $\alpha \in [0, 1]$ is Proposition 19 at $p = 1 + r$, whose proof is dimension-free; achievability of the frontier point is (ii).

For $r = 1$ the bijection $c = W^{-\top}(a, b)^\top$ carries (Q_0, Q_1, h) and the constraint to their Section IV forms, so the two weighted programs are identical with the same unique minimizer, and the two stationarity systems are equivalent parameterizations of it. At $\alpha = 0$, (42) reads $c = (1 - \gamma_1)[(1 - \gamma_1)I + \gamma_1\Lambda]^{-1}y_0$; substituting into $\nu = D - |y_0 - c|^2$ and $\gamma_1 = \nu/(c^\top\Lambda c + \nu)$ leaves one scalar equation in $\gamma_1 \in (0, 1)$, the vector-context generalization of the quadratic (20). $\square$

VII. THE BINARY FRONTIER

The binary case is in this paper for two reasons: it shows that the misalignment phenomenon and the strict rate–content tradeoff are not Gaussian artifacts, the frontier being fully characterized here too (Proposition 28), and it shows the functional is explicitly computable outside the Gaussian family. The conditional content itself coincides with a known common-reconstruction rate (Appendix C); the frontier is traced here, unlike its Gaussian counterpart, by an elementary self-contained argument.

Let (Y, V) be a doubly symmetric binary pair (fair marginals, $\Pr[Y \neq V] = p \in (0, \frac{1}{2})$), context V , side information $S = V \oplus W$ with $W \sim \text{Bern}(q)$, $q \in [0, \frac{1}{2}]$, independent. The description is the reproduction $\hat{Y} \in \{0, 1\}$ itself (richer descriptions only cost more: for any stored Z from which the decoder computes $\hat{Y} = f(Z)$, $I(Y, V; Z | S) \geq I(Y, V; \hat{Y} | S)$ by the data-processing inequality), and the distortion is Hamming, $\Pr[\hat{Y} \neq Y] \leq D$. Write $x * q = x(1 - q) + (1 - x)q$, $\ell(x) = \ln \frac{1-x}{x}$, and h_2 for the binary entropy function in bits.

Lemma 25. $L(C) := I(Y, V; \hat{Y} | S)$ is convex in the channel $C = P(\hat{y} | y, v)$, and the Hamming distortion is linear in C .

Proof. The Markov chain $\hat{Y} - (Y, V) - S$ means that one channel C is shared across the values of S , so $L(C) = \sum_s P(s) I_s(C)$, where $I_s(C)$ is the mutual information of the channel C under the input law $P(Y, V | S = s)$. The input laws depend on s , but each I_s is convex in C at its own fixed input law, so the nonnegative combination is convex in C . Linearity of the distortion is immediate. $\square$

Lemma 26. Fix $p \in (0, \frac{1}{2})$, $q \in (0, \frac{1}{2}]$, $D \in (0, \frac{1}{2})$. Parameterize the constraint segment $(1 - p)d_0 + pd_1 = D$, $(d_0, d_1) \in (0, 1)^2$, by $d_0 \in (\max\{0, \frac{D-p}{1-p}\}, \frac{D}{1-p})$, an open subinterval of $(0, 1)$, and define the tilt residual

$$\psi(d_0) := \ell(d_0) - \ell(d_1(d_0)) - 2(1 - 2q)\ell(u(d_0)),$$

$$d_1(d_0) = \frac{D - (1 - p)d_0}{p}, \quad u = a * q, \quad a = 2(1 - p)d_0 + p - D.$$

Then ψ has exactly one zero on the open segment, and the family value $L(d_0)$ attains its minimum over the closed segment exactly there. The same statements hold at $q = 0$ when $D < p$.

Proof. Work in nats, with h_e the binary entropy function in nats, so the L of Theorem 27 in bits is this L divided by $\ln 2$: $L(d_0) = h_e(u) - (1-p)h_e(d_0) - ph_e(d_1(d_0))$. Differentiating with $h'_e(x) = \ell(x)$, $u'(d_0) = 2(1-p)(1-2q)$, and $d'_1(d_0) = -(1-p)/p$,

$$L'(d_0) = 2(1-p)(1-2q)\ell(u) - (1-p)\ell(d_0) + (1-p)\ell(d_1) = -(1-p)\psi(d_0).$$

Existence and interiority. We first record the boundary behavior. At the left endpoint: if $D \leq p$ the endpoint is $d_0 = 0$ and $\ell(d_0) \rightarrow +\infty$; if additionally $D = p$ then also $d_1 \rightarrow 1$ and $-\ell(d_1) \rightarrow +\infty$, the same sign; if $D > p$ the endpoint is $d_0 = (D-p)/(1-p) > 0$ with $d_1 \rightarrow 1$, so $-\ell(d_1) \rightarrow +\infty$ with the other terms bounded. The u term stays bounded in every case: $u = a * q \in [q, 1-q]$ with $q > 0$. Hence $\psi \rightarrow +\infty$ at the left endpoint. At the right endpoint $d_0 = D/(1-p) < 1$ (since $1-p > \frac{1}{2} > D$): $d_1 \rightarrow 0$, so $-\ell(d_1) \rightarrow -\infty$ with the other terms bounded, and $\psi \rightarrow -\infty$. ψ is continuous on the open segment, so it has a zero there by the intermediate value theorem, and trivially every zero is interior.

Zeros are minimizers. The family channel is affine in (d_0, d_1) , hence affine in d_0 along the segment, and L is convex in the channel (Lemma 25), so L is convex on the segment. For a convex differentiable function, $L'(d_0) = 0$ at an interior point makes that point a global minimizer; conversely $L' \rightarrow -\infty$ at the left endpoint and $L' \rightarrow +\infty$ at the right (the sign flips of ψ), so the minimum of L over the closed segment is attained in the interior and only at zeros of ψ .

Uniqueness. The minimizer set of the convex L is a closed interval on which L is constant. Suppose it contains two distinct points. Then $L'' \equiv 0$ on a nondegenerate subinterval. But L is real analytic on the open segment: d_1 and a are affine in d_0 with values in $(0, 1)$, $u = a(1-2q) + q \in (0, 1)$, and h_e is real analytic on $(0, 1)$. By the identity theorem, $L'' \equiv 0$ on the whole connected open segment, so L' is constant there, contradicting $L' \rightarrow -\infty$ at the left endpoint. Hence the minimizer is unique, and since the zeros of ψ are exactly the interior stationary points of L , which are exactly its minimizers, ψ has exactly one zero.

At $q = 0$ with $D < p$ the same argument applies verbatim: the left endpoint is $d_0 = 0$ with $d_1(0) = D/p < 1$, and $u = a \in (p-D, p+D) \subset (0, 1)$ stays bounded away from 0 and 1, so the boundary blow-ups and the analyticity are as before. $\square$

Fig. 4 shows the objective, the tilt residual, and the rate-content frontier at representative instances.

Theorem 27. For $D \in (0, \frac{1}{2})$, the conditional coordinate $L(D) = \min I(Y, V; \hat{Y} | S)$ over channels $P(\hat{y} | y, v)$ (Markov $\hat{Y} - (Y, V) - S$) with $\Pr[\hat{Y} \neq Y] \leq D$ is attained in the two-parameter symmetric family $\hat{Y} = Y \oplus E$, $\Pr[E = 1 | Y \oplus V = j] = d_j$, and equals

$$L(D) = h_2(u) - (1-p)h_2(d_0) - ph_2(d_1), \quad u = a * q, \quad a = (1-p)d_0 + p(1-d_1),$$

where, for $q \in (0, \frac{1}{2}]$ with any $D \in (0, \frac{1}{2})$, and for $q = 0$ with $D < p$, the pair (d_0, d_1) is the unique solution, interior to the constraint segment (Lemma 26), of the active constraint $(1-p)d_0 + pd_1 = D$ together with the tilt equation

$$\ell(d_0) - \ell(d_1) = 2(1-2q)\ell(u).$$

At the boundary point $q = 0$, $D = p$, the minimum $L = 0$ is attained at the segment endpoint $(d_0, d_1) = (0, 1)$, the reproduction $\hat{Y} = V$. (In the excluded sliver $q = 0$, $D > p$, the constraint is inactive and $L(D) = 0$, attained non-uniquely by reproductions measurable in V alone; the tilt system still has a root there, the degenerate point $d_1 = 1 - d_0$, which evaluates to the correct value 0. For $q > 0$ the constraint is active on all of $(0, \frac{1}{2})$: L is convex, nonincreasing, positive by the last claim below, and 0 at $D = \frac{1}{2}$, hence strictly decreasing.) On the same channel the rate coordinate is $R = 1 - (1-p)h_2(d_0) - ph_2(d_1)$ and the discount is exactly $R - L = 1 - h_2(u) = I(\hat{Y}; S)$ (the chain-rule identity $I(Y, V; \hat{Y}) - I(Y, V; \hat{Y} | S) = I(\hat{Y}; S)$, valid under the Markov chain, in closed form: $a = \Pr[\hat{Y} \neq V]$, $u = \Pr[\hat{Y} \neq S]$). Anchors: at $q = 0$ the system gives $a = (p-D)/(1-2D)$ and $L = h_2(p) - h_2(D)$ (Gray's conditional rate-distortion function [4], for $D \leq p$); at $q = \frac{1}{2}$ it gives $d_0 = d_1 = D$ and $L = 1 - h_2(D)$ (the marginal rate-distortion function; the context is useless exactly when the side information is). For $q \in (0, \frac{1}{2}]$ and $D < \frac{1}{2}$, $L(D) > 0$.

Proof. Reduction and convexity. Binary $\hat{Y}$ is definitional (DPI above); convexity of L in the shared channel and linearity of the distortion are Lemma 25.

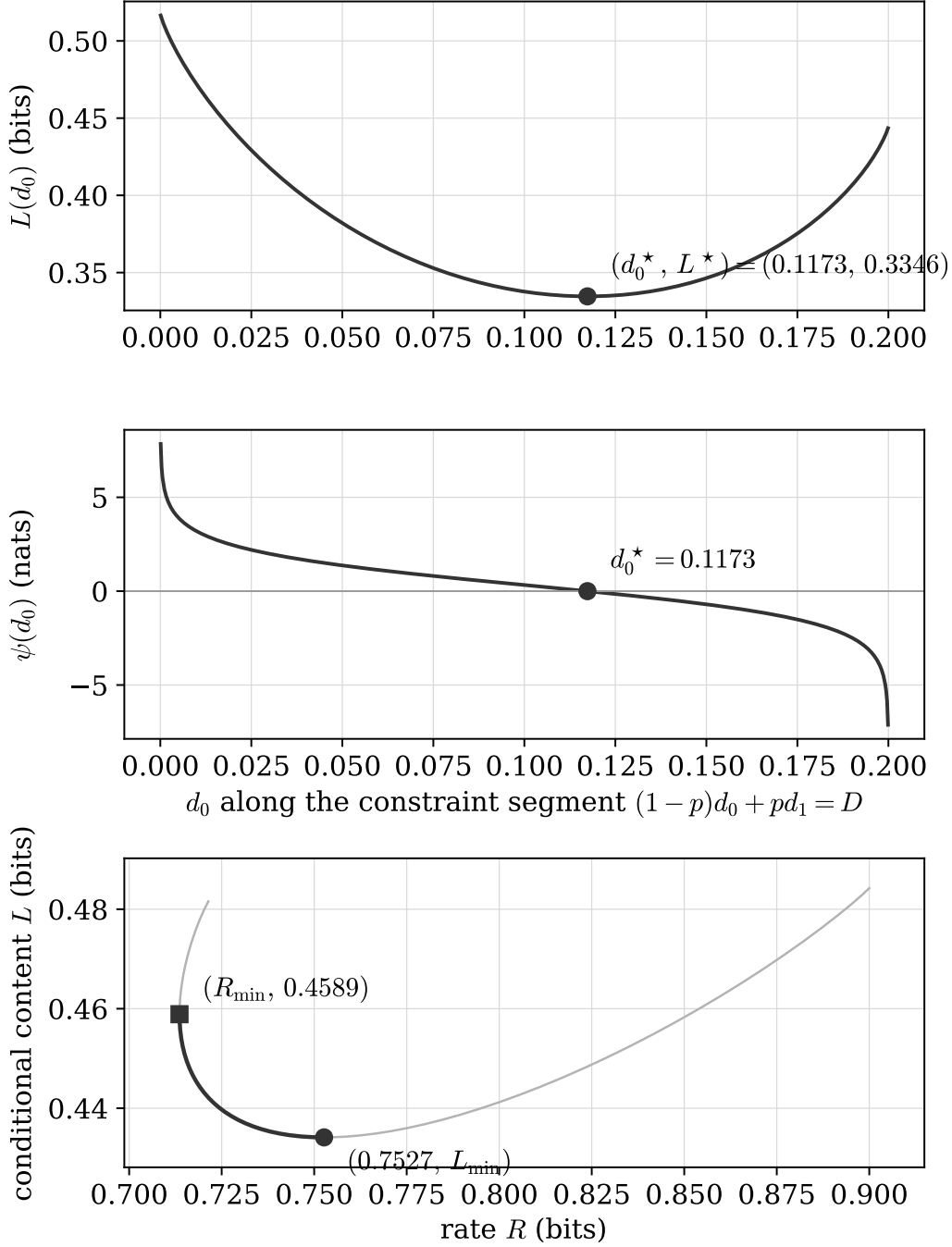


Fig. 4. The binary objective $L(d_0)$ (top) and the tilt residual $\psi(d_0)$ (middle) along the constraint segment $(1-p)d_0 + pd_1 = D$ at $(p, q, D) = (0.25, 0.15, 0.15)$: the residual crosses zero exactly once, at $d_0^* = 0.1173$ ($d_1^* = 0.2480$), and L attains its unique minimum 0.3346 bits there, as Lemma 26 asserts. Bottom: the binary rate–content frontier at $(p, q, D) = (0.1, 0.1, 0.05)$, the curve $(R(d_0), L(d_0))$ along the segment (Proposition 28), with the Pareto arc from the tilt root $d_0^* = 0.0282$ (the L -minimizer, $L_{\min} = 0.4341$ bits) to $d_0 = D$ (the R -minimizer, $R_{\min} = 1 - h_2(D) = 0.7136$ bits) drawn bold. The tradeoff is strict: the L -optimal point pays 0.0391 bits of excess rate and the R -optimal point pays 0.0248 bits of excess conditional content.

Symmetrization. The global bit flip $\sigma : (y, v, s, \hat{y}) \mapsto (1-y, 1-v, 1-s, 1-\hat{y})$ preserves the joint law of (Y, V, S) : the pair (Y, V) is doubly symmetric with fair marginals, so $(1-Y, 1-V) \stackrel{d}{=} (Y, V)$, and $W \sim \text{Bern}(q)$ is independent of (Y, V) , so $S = V \oplus W$ flips with V . The flip also preserves the distortion, and $L(C^\sigma) = L(C)$; averaging C with C^σ preserves distortion and, by convexity, does not increase L . So an optimum exists among σ -symmetric

channels, and these are exactly the family $\{(d_0, d_1)\}$: σ -symmetry forces $\Pr[\hat{Y} \neq Y \mid y, v]$ to depend only on $y \oplus v$.

Closed form. On the family, $H(\hat{Y} \mid Y, V) = (1-p)h_2(d_0) + ph_2(d_1)$, and $\hat{Y} \oplus S = J \oplus E \oplus W$ ($J = Y \oplus V$) with $\Pr[J \oplus E = 1] = a$. The step $H(\hat{Y} \mid S) = h_2(a * q)$ needs $\hat{Y} \oplus S \perp S$, which holds because the joint law of $(\hat{Y}, S)$ is σ -invariant (σ -invariant source, σ -symmetric channel), forcing equal conditional flip probabilities given either value of S ; the same symmetry gives $H(\hat{Y}) = 1$, hence R .

Tilt equation. Substituting $d_1 = (D - (1-p)d_0)/p$ makes $a = 2(1-p)d_0 + p - D$ affine in d_0 ; differentiating L and using $h_2'(x) = \ell(x)/\ln 2$ and dropping the common factor $1/\ln 2$ gives $(1-p)[2(1-2q)\ell(u) - \ell(d_0) + \ell(d_1)] = 0$. Since L restricted to the affine family slice is convex (restriction of a convex function), interior stationarity is sufficient, not merely necessary.

Anchors and positivity. $q = \frac{1}{2}$: $u = \frac{1}{2}$, the tilt equation forces $d_0 = d_1 = D$. $q = 0$: substitution of Gray's backward channel; equality of values is elementary algebra. Positivity, in two branches over the family (which suffices by attainment): if $a \neq \frac{1}{2}$, then $h_2(u) > h_2(a) \geq (1-p)h_2(d_0) + ph_2(1-d_1) = (1-p)h_2(d_0) + ph_2(d_1)$, where the strict step holds because $u = a * q$ is strictly closer to $\frac{1}{2}$ for $q \in (0, \frac{1}{2}]$ and the second step by concavity; so $L > 0$. If $a = \frac{1}{2}$, then $u = \frac{1}{2}$, $h_2(u) = 1$, and $L = 1 - (1-p)h_2(d_0) - ph_2(d_1) \geq 1 - h_2(D) > 0$ by concavity at the distortion. $\square$

Reproduction alphabets larger than binary are already covered by the data-processing reduction that opens this section; no separate argument is needed for them. The unconditional Lagrangian bounds, which assume no convexity, numerically re-confirm that reduction and the family-plus-tilt optimum.

Proposition 28. *The symmetric family exhausts not only the conditional coordinate but the entire rate–content frontier at distortion D : the Pareto frontier of the binary rate–content region is the arc $d_0 \in [d_0^*, D]$ of the curve*

$$d_0 \mapsto (R(d_0), L(d_0)), \quad R(d_0) = 1 - (1-p)h_2(d_0) - ph_2(d_1(d_0)),$$

along the constraint segment. On the segment R is uniquely minimized at $d_0 = d_1 = D$, where $R = 1 - h_2(D)$ is the marginal rate–distortion function, and L at the tilt root d_0^ of Lemma 26. For $q \in (0, \frac{1}{2})$ the two minimizers are distinct, $d_0^* < D$, and $[d_0^*, D]$ is a strict tradeoff arc, L increasing and R decreasing, with both coordinates worsening outside it; at $q = \frac{1}{2}$ the frontier is a single corner.*

Proof. The rate coordinate $R = I(Y, V; \hat{Y})$ is convex in the shared channel by the argument of Lemma 25 with the trivial conditioner, so every weighted objective $\alpha R + (1-\alpha)L$, $\alpha \in [0, 1]$, is convex in the channel. The global bit flip σ of Theorem 27's proof preserves the joint law of (Y, V, S) , the distortion, and both coordinates: the flip is a law-preserving bijective relabeling of both arguments of each mutual information, so $R(C^\sigma) = R(C)$ and $L(C^\sigma) = L(C)$. Averaging a channel with its flip therefore preserves the distortion and, by convexity, does not increase any weighted objective, and larger reproduction alphabets are covered by the data-processing reduction in both coordinates. Every weighted optimum is thus attained in the family $\{(d_0, d_1)\}$, and the distortion constraint may be taken active there: both coordinates are nonnegative and vanish at $d_0 = d_1 = \frac{1}{2}$, so by convexity the objective does not increase along the straight path from any point with distortion slack toward $(\frac{1}{2}, \frac{1}{2})$, while the distortion, affine in (d_0, d_1) with positive weights, rises continuously to $\frac{1}{2} > D$ and meets the budget on the way. Hence the frontier lies on the displayed curve along the segment, and the arc between the two minimizers identified next is the Pareto-optimal part. On the segment, R is strictly convex with its unique minimum at $d_0 = d_1 = D$ (strict concavity of $(1-p)h_2(d_0) + ph_2(d_1)$ and the stationarity $\ell(d_0) = \ell(d_1)$), where $R = 1 - h_2(D)$, the marginal rate–distortion function; L has its unique minimum at the tilt root d_0^* (Lemma 26). For $q \in (0, \frac{1}{2})$ the two minimizers are distinct, since $\psi(D) = -2(1-2q)\ell(u(D)) < 0$ (there $u(D) < \frac{1}{2}$ because $a(D) = p + D(1-2p) < \frac{1}{2}$), so $d_0^* < D$: on $[d_0^*, D]$ the curve is a strict tradeoff arc, L increasing and R decreasing, and outside that arc both coordinates worsen. At $q = \frac{1}{2}$ the two minimizers coincide at $d_0 = D$ and the frontier is a single corner. Fig. 4 traces the frontier at a representative instance. $\square$

The binary conditional-content function of this paper coincides with the common-reconstruction rate of Lu, Xu, Zhang, Feng, and Wang [21] (with the Heegard–Berger companion [22]), whose encoder observes side information that may differ from the decoder's. Under the relabeling of Appendix C their function is identically $L(D)$, so the binary $L(D)$ here is theirs; the contribution of this section is the joint (R, L) frontier they do not treat. Two further common-reconstruction calculations are decoder-side-information ones. Steinberg's binary example [11] evaluates the constrained functional for a binary symmetric pair: the decoder's side information is the source through a

binary symmetric channel, the distortion is Hamming, the optimal reverse test channel is a BSC of crossover D independent of the pair correlation, and the value is $h_2(\rho_0 * D) - h_2(D)$ at side-channel crossover ρ_0 . Ahmadi, Tandon, Simeone, and Poor [13] computed common-reconstruction functions for a binary source with *erased* side information under Hamming and erasure distortion, in point-to-point, Heegard–Berger, and cascade configurations; the point-to-point value is $p_e(1 - h_2(D))$ at erasure rate p_e . Both computations live in the decoder-side-information setting: S sits at the decoder, the encoder observes the source alone, and the common-reconstruction constraint disciplines how the decoder may use S . Theorem 27 differs structurally. Here S is held by a third party that never decodes, the encoder observes the pair (Y, V) , and the constraint binds the unconditional Hamming error $\Pr[\hat{Y} \neq Y]$. The function is anchored at Gray’s conditional function at $q = 0$ and at the marginal rate–distortion function at $q = \frac{1}{2}$, an interpolation with no counterpart in the decoder-side calculations. Its $q = 0$ endpoint is also the common-reconstruction value of [21] at $p_2 = 0$ (Appendix C). An encoder restricted to Y alone faces exactly the constrained problem of the pair (Y, S) , a binary symmetric pair with crossover $p * q$, and pays Steinberg’s value there (the marginalization identity of Section IV is alphabet-free); the tilt equation is the price of pair access.

VIII. DISCUSSION AND CONCLUSION

This paper characterized the minimal conditional content of a description whose encoder observes the pair (Y, V) while the conditioner S stays with a third party: the closed form of Theorem 15(a) for Y and V scalar linear functionals of an arbitrary-dimensional jointly Gaussian source, the exact rate–content region of Theorem 15(b) with its two-water-level frontier and the misalignment dichotomy of Corollary 20, the attainment condition for the conditional-variance bound (Remark 23), the vector-context extension of Theorem 24, and the binary rate–content frontier of Theorem 27 and Proposition 28. The function degenerates to the classical rate–distortion function at $\rho = 0$ and to Gray’s conditional function as $\tau^2 \rightarrow 0$, and meets Steinberg’s scalar Gaussian formula at the $\rho^2 \rightarrow 1$ boundary. The structural conclusion is Corollary 22: the function is not determined by the reduced marginal (Y, S) but by what the encoder can see, so conditional description content is a property of the encoding configuration, not of the source and side-information pair alone.

Four problems are open. The exact rate–content region for a joint description serving two described variables is not settled; its stationarity system is known in necessary-condition form only. The analogous function for a described variable of dimension greater than one is open. Stationary sources, and the process and spectral versions of the function, are not treated here. The finite-blocklength behavior of the pair (R_n, L_n) is unexplored, and the content coordinate is not a routine second-order corollary of the rate one: the dispersion of L_n is governed by the variance of a conditional information density, which the quantize-and-pass-to-the-limit argument of Appendix A does not control, so a joint (R_n, L_n) dispersion region does not follow from the rate dispersion alone.

Everything in this paper is single-letter and memoryless, Gaussian and binary. Every closed form is verified by the independent numerical checks of Appendix B.

APPENDIX A

THE GAUSSIAN OPERATIONAL THEOREM

This appendix proves the operational region theorem for the jointly Gaussian source of Section III, replacing the customary appeal to an unspecified abstract-alphabet extension. The source letter is $X = (Y, V)$, jointly Gaussian, the pair T of Section III taken as the source letter, which Lemma 11 permits; the conditioner is $S \in \mathbb{R}^r$ as in Section III ($r = 1$) or Section VI; the distortion is $d(x, \hat{y}) = (y - \hat{y})^2$, unbounded; codes, R_n , D_n , and $L_n = H(M | S^n)/n$ are as in Section II, and achievability is Definition 1. The proof establishes, in order: the converse with differential entropies; closedness and right-continuity of the union; restriction to Gaussian channels; and achievability in three steps, quantizing the conditioner (Step A), then the source and the reproduction (Step B, with a finite reproduction alphabet at every stage and the Markov chain preserved by construction), then returning to the unquantized conditioner through $H(M | S^n) \leq H(M | S_\Delta^n)$ (Step C). The rate and content coordinates converge by data processing, martingale convergence, and lower semicontinuity of mutual information; the distortion is transferred exactly by memorylessness rather than by weak convergence.

One device is fixed first, the *saturating dyadic quantizers*: for $k \geq 1$, q_k partitions $\mathbb{R}$ into the bounded cells $[j2^{-k}, (j+1)2^{-k})$, $-4^k \leq j < 4^k$, and the two saturating tail cells $(-\infty, -2^k)$ and $[2^k, \infty)$, and maps each point of a bounded cell to the cell’s left endpoint and the two tail cells to -2^k and 2^k , so that $|q_k(u) - u| \leq 2^{-k} + |u| \mathbf{1}\{|u| > 2^k\}$

for every real u . A vector is quantized coordinatewise, so its cells are rectangles. The partitions are nested, every cell edge of q_k being an edge of q_{k+1} , so for any random vector Z the σ -fields $\sigma(q_k(Z))$ increase with k , and their union generates $\sigma(Z)$, since the dyadic rectangles generate the Borel σ -field. Write $S_k := q_k(S)$.

Theorem 29. *For the Gaussian source under the standing assumptions of Section III ($\tau^2 > 0$, $\Sigma_T \succ 0$), with the pair $T = (Y, V)$ taken as the source letter X (Lemma 11), and every $0 < D < 1$,*

$$\mathcal{RW}(D) = \bigcup_{p(\hat{y}|x): \mathbb{E}(Y - \hat{Y})^2 \leq D} \{(R, L) : R \geq I(X; \hat{Y}), L \geq I(X; \hat{Y} | S)\},$$

and the union is unchanged when restricted to jointly Gaussian test channels.

The compactness step of the proof uses $D < 1$. For $D \geq 1$ the statement follows from the zero reproduction, which meets the distortion constraint with $R = L = 0$, and $D = 0$ is the extended-real limit with both coordinates diverging. Throughout, L_n is the coding quantity (6); no step of the proof depends on Remark 8.

Proof. Converse. The chain of Theorem 6's converse (the discrete operational region theorem) holds verbatim with differential entropies for the continuous source. M is discrete, so $\log |\mathcal{M}_n| \geq H(M) \geq I(X^n; M) \geq I(X^n; \hat{Y}^n)$ and $H(M | S^n) \geq I(X^n; M | S^n) \geq I(X^n; \hat{Y}^n | S^n)$ are unchanged. The single-letterization holds with differential entropy h in place of H : for the jointly Gaussian source all the differential entropies involved are finite (the unconditional ones are Gaussian values, and each conditional one differs from an unconditional one by a mutual information pinned between 0 and $H(M) \leq \log |\mathcal{M}_n| < \infty$, since $\hat{Y}^n$ is a function of the finite-valued M), the i.i.d. chain rules are unchanged, and conditioning reduces differential entropy. The per-letter induced channels satisfy the Markov constraint by the i.i.d. structure, exactly as in the finite case; the mixture channel $\bar{q}$ is defined by averaging the induced conditional laws, its distortion is D_n , and the convexity step is Lemma 3 in the general-alphabet form of Remark 4. Hence (R_n, L_n) lies in the displayed union at distortion D_n .

Closedness and the passage from D_n to D . By the restriction step below, the union at any distortion level $D' \in (0, 1)$ equals the union over jointly Gaussian test channels, and these are parameterized by their regression vector and residual variance: in the whitened coordinates of (37) (the simultaneous congruence $\Sigma_T \rightarrow I$, $\Sigma_{T|S} \rightarrow \Lambda$), by pairs $\theta = (c, \nu)$ with $Q_0 = |c|^2$, $Q_1 = c^\top \Lambda c$, distortion $|y_0 - c|^2 + \nu \leq D'$, and coordinates $(\frac{1}{2} \log_2 \frac{Q_0 + \nu}{\nu}, \frac{1}{2} \log_2 \frac{Q_1 + \nu}{\nu})$. The distortion constraint confines θ to a compact set: $|y_0 - c|^2 \leq D'$ is a closed ball and $0 \leq \nu \leq D'$; and feasibility forces $c \neq 0$ there, since $c = 0$ gives distortion at least $|y_0|^2 = 1 > D'$. Both coordinates are lower semicontinuous on that set with values in $(0, \infty]$, finite and continuous where $\nu > 0$ and diverging as $\nu \rightarrow 0$ at $c \neq 0$. Now let (R_j, L_j) lie in the union at levels D'_j with $\limsup_j D'_j \leq D$ and $(R_j, L_j) \rightarrow (R, L)$ finite. Pick feasible θ_j whose quadrant contains (R_j, L_j) ; the θ_j eventually lie in the compact set at level $D + \delta$ for each $\delta > 0$, so a subsequence converges to some θ^* , feasible at D by continuity of the distortion in θ . Lower semicontinuity gives $\frac{1}{2} \log_2 [(Q_0 + \nu)/\nu](\theta^*) \leq \liminf_j \frac{1}{2} \log_2 [(Q_0 + \nu)/\nu](\theta_j) \leq R$, and likewise for the conditional coordinate, so (R, L) lies in θ^* 's quadrant, hence in the union at D . This establishes both closedness at fixed distortion and right-continuity in the distortion level, and with $\limsup D_n \leq D$ places every limit point of (R_n, L_n) in the union at D .

Restriction to Gaussian channels. For any admissible channel, Lemma 13 (Gaussian exhaustion: the two determinant lower bounds, met with equality by the jointly Gaussian channel matching the second moments) produces the jointly Gaussian channel with the same second moments and the same distortion, whose coordinates $I(X; \hat{Y})$ and $I(X; \hat{Y} | S)$ are componentwise no larger than the original channel's. Its quadrant therefore contains the original channel's quadrant, so the union over all channels equals the union over jointly Gaussian channels. It thus suffices to prove achievability for a jointly Gaussian test channel.

Achievability, Step A: quantizing the conditioner. Fix a jointly Gaussian admissible channel $p(\hat{y} | x)$ with distortion at most D and residual variance $\nu > 0$ (a channel with $\nu = 0$ has both coordinates infinite, by the closedness step, and contributes nothing to the union), and fix $\epsilon > 0$. Along the saturating dyadic quantizers,

$$I(\hat{Y}; S_k) \uparrow I(\hat{Y}; S) \quad (k \rightarrow \infty) : \quad (43)$$

the sequence is nondecreasing because the partitions are nested (data processing), and its supremum is $I(\hat{Y}; S)$ by monotone convergence of mutual information under σ -fields increasing to the Borel σ -field of S [6, Ch. 2]: mutual information is the supremum of the mutual informations of finite measurable partitions of the two sample spaces,

and every finite Borel partition of the S -space is approximated, up to sets of arbitrarily small measure, by partitions measurable with respect to some $\sigma(S_k)$. Choose k with

$$I(\hat{Y}; S_\Delta) \geq I(\hat{Y}; S) - \epsilon, \quad S_\Delta := S_k. \quad (44)$$

S_Δ is a function of S , so the Markov chain $\hat{Y} - X - S_\Delta$ holds, and by the chain-rule identity used throughout ($I(X; \hat{Y}) - I(X; \hat{Y} | S') = I(\hat{Y}; S')$ for any conditioner S' with $\hat{Y} - X - S'$),

$$I(X; \hat{Y} | S_\Delta) = I(X; \hat{Y}) - I(\hat{Y}; S_\Delta) \leq I(X; \hat{Y} | S) + \epsilon. \quad (45)$$

Achievability, Step B: discretizing the source and the reproduction. The pair source is now (X, S_Δ) with continuous X and finite S_Δ , and the test channel is the jointly Gaussian $p(\hat{y} | x)$. Theorem 6 requires finite alphabets and a bounded distortion, so we discretize both and pass to the limit. For $m \geq 1$ let $X_m := q_m(X)$, the coordinatewise quantizer applied to $X = (Y, V)$, and $\hat{Y}_m := q_m(\hat{Y})$, so that $|\hat{Y}_m - \hat{Y}| \leq 2^{-m} + |\hat{Y}| \mathbf{1}\{|\hat{Y}| > 2^m\}$ and $\hat{Y}_m \rightarrow \hat{Y}$ in L^2 . The finite-alphabet surrogate consists of the source X_m and the side information S_Δ under their true joint law, which is that of a discrete memoryless pair because (X_m^n, S_Δ^n) is a coordinatewise function of the i.i.d. pairs (X^n, S^n) ; the finite reproduction alphabet $q_m(\mathbb{R}) \subset \mathbb{R}$; the test channel

$$p_m(\hat{y} | x_m) := \Pr(\hat{Y}_m = \hat{y} | X_m = x_m),$$

the conditional law of the quantized reproduction given the quantized source; and the surrogate distortion

$$d_m(x_m, \hat{y}) := \mathbb{E}[(Y - \hat{y})^2 | X_m = x_m] = \text{Var}(Y | X_m = x_m) + (\mathbb{E}[Y | X_m = x_m] - \hat{y})^2,$$

which is bounded, since both alphabets are finite. Under the surrogate joint law $p(x_m, s_\Delta) p_m(\hat{y} | x_m)$ the Markov chain $\hat{Y}_m - X_m - S_\Delta$ holds by construction, so Lemma 7 applies to the surrogate and, for every $\epsilon > 0$ and every m , supplies a deterministic code, a function of X_m^n alone, with $|\mathcal{M}_n| \leq 2^{n(I(X_m; \hat{Y}_m) + 2\epsilon)}$, surrogate distortion at most $\mathbb{E}_m d_m + \epsilon + 2\epsilon_n d_{\max}(m)$, and

$$H(M | S_\Delta^n) \leq n(I_m(X_m; \hat{Y}_m | S_\Delta) + 3\epsilon) + 1 + 2\epsilon_n n(I(X_m; \hat{Y}_m) + 2\epsilon), \quad (46)$$

where $\mathbb{E}_m$ and I_m are computed under the surrogate joint law, $d_{\max}(m) = \max_{x_m, \hat{y}} d_m(x_m, \hat{y})$ is finite for each m but grows with m because the tail cells sit at $\pm 2^m$, and $\epsilon_n \rightarrow 0$ for each fixed (ϵ, k, m) . The limits below are taken in this order: ϵ is fixed; k is fixed by Step A; m is fixed by (ii) and (iii); then $n \rightarrow \infty$, so that $2\epsilon_n d_{\max}(m) = o(1)$ and $2\epsilon_n n(\cdot) = o(n)$, which is how they are written below; and finally $\epsilon \downarrow 0$ along a diagonal sequence. Also, the surrogate law of the pair $(X_m, \hat{Y}_m)$ is their true joint law, so $I(X_m; \hat{Y}_m)$ needs no subscript. Three facts transfer these guarantees to the original problem.

(i) *The code is a code for the original source, with the same distortion.* X_m^n is a function of X^n , so the code is admissible for X^n . Each reproduction letter $\hat{y}_i$ is a function of X_m^n , and given X_m^n the conditional law of Y_i depends on $X_{m,i}$ alone, by memorylessness; hence $\mathbb{E}[(Y_i - \hat{y}_i)^2 | X_m^n] = d_m(X_{m,i}, \hat{y}_i)$, and taking expectations and averaging over i identifies the true distortion with the surrogate distortion.

(ii) *The surrogate distortion converges to the original one.* Since the surrogate law of $(X_m, \hat{Y}_m)$ is the true one,

$$\mathbb{E}_m d_m(X_m, \hat{Y}_m) = \mathbb{E}[\text{Var}(Y | X_m)] + \mathbb{E}[(\mathbb{E}[Y | X_m] - \hat{Y}_m)^2] \rightarrow \mathbb{E}(Y - \hat{Y})^2 \leq D \quad (m \rightarrow \infty),$$

because the nested partitions generate $\sigma(X)$, so $\mathbb{E}[Y | X_m] \rightarrow Y$ in L^2 by the martingale convergence theorem, which sends the first term to zero, and $\hat{Y}_m \rightarrow \hat{Y}$ in L^2 . (The surrogate distortion is not $\mathbb{E}(Y - \hat{Y}_m)^2$ under the true law, since Y and $\hat{Y}_m$ are dependent given X_m ; the argument does not use that identity.)

(iii) *The surrogate coordinates approach the original ones.* The rate obeys $I(X_m; \hat{Y}_m) \leq I(X; \hat{Y})$ by data processing. For the conditional coordinate, the chain-rule identity under the surrogate Markov chain gives $I_m(X_m; \hat{Y}_m | S_\Delta) = I(X_m; \hat{Y}_m) - I_m(\hat{Y}_m; S_\Delta) \leq I(X; \hat{Y}) - I_m(\hat{Y}_m; S_\Delta)$, so it remains to show that $\liminf_m I_m(\hat{Y}_m; S_\Delta) \geq I(\hat{Y}; S_\Delta)$. The surrogate law of $(\hat{Y}_m, S_\Delta)$ is $Q_m(B \times \{j\}) = \mathbb{E}[\Pr(\hat{Y}_m \in B | X_m) \Pr(S_\Delta = j | X_m)]$, and for every bounded function f on $\mathbb{R} \times q_k(\mathbb{R}^r)$, continuous in its first argument,

$$\begin{aligned} \int f dQ_m &= \mathbb{E} \left[\sum_j \mathbb{E}[f(\hat{Y}_m, j) | X_m] \Pr(S_\Delta = j | X_m) \right] \\ &\rightarrow \mathbb{E} \left[\sum_j \mathbb{E}[f(\hat{Y}, j) | X] \Pr(S_\Delta = j | X) \right] = \mathbb{E}f(\hat{Y}, S_\Delta) : \end{aligned}$$

$f(\hat{Y}_m, j) \rightarrow f(\hat{Y}, j)$ boundedly, the conditional expectations given X_m converge to those given X by the martingale convergence theorem, and the last equality is the conditional independence of $\hat{Y}$ and S_Δ given X , the Markov chain $\hat{Y} - X - S$. So Q_m converges weakly to the true law of $(\hat{Y}, S_\Delta)$, and mutual information is lower semicontinuous under weak convergence of the joint law [7]. Hence for all large m , $I_m(\hat{Y}_m; S_\Delta) \geq I(\hat{Y}; S_\Delta) - \epsilon$, and with (45),

$$I_m(X_m; \hat{Y}_m | S_\Delta) \leq I(X; \hat{Y}) - I(\hat{Y}; S_\Delta) + \epsilon = I(X; \hat{Y} | S_\Delta) + \epsilon \leq I(X; \hat{Y} | S) + 2\epsilon.$$

Fix m so large that (ii) gives $\mathbb{E}_m d_m \leq D + \epsilon$ and (iii) holds. The selected code then has rate at most $I(X; \hat{Y}) + 2\epsilon$, true distortion at most $D + 2\epsilon + o(1)$ by (i), and, by (46) and (iii),

$$H(M | S_\Delta^n) \leq n(I(X; \hat{Y} | S) + 5\epsilon + o(1)). \quad (47)$$

Achievability, Step C: back to the unquantized conditioner. S_Δ^n is a function of S^n , so $\sigma(S_\Delta^n) \subseteq \sigma(S^n)$ and conditioning on the finer σ -algebra cannot increase entropy:

$$H(M | S^n) \leq H(M | S_\Delta^n) \leq n(I(X; \hat{Y} | S) + 5\epsilon + o(1)), \quad (48)$$

by (47).

Collect. For every $\epsilon > 0$ the triple $(I(X; \hat{Y}) + 2\epsilon, I(X; \hat{Y} | S) + 5\epsilon, D + 2\epsilon)$ is achievable; a diagonal sequence $\epsilon_j \downarrow 0$ of codes achieves, in the lim sup sense of Definition 1, the corner $(I(X; \hat{Y}), I(X; \hat{Y} | S))$ at distortion D , and the quadrant above the corner is immediate from the definition. Together with the converse and the restriction step, this proves the theorem. $\square$

The imports are stated for the record: monotone convergence of mutual information under refining partitions [6, Ch. 2] and its lower semicontinuity under weak convergence [7]. Every other step is Theorem 6 or is proved here; no abstract-alphabet covering or Markov lemma is invoked.

APPENDIX B NUMERICAL VERIFICATION

Each closed form and each displayed matrix identity was checked independently of the derivations, symbolically by exact computer algebra and numerically against direct optimization over test-channel moments, twice over: by the author's harness, written alongside the proofs, and by a separately commissioned re-derivation, archived with the repository. All checks pass. The core scalar algebra is additionally machine-checked in two further engines, a Lean 4/Mathlib module built clean with zero `sorry` and the MATLAB Symbolic toolbox. The matrix-general exhaustion lemma, the measure-theoretic steps (single-letterization, covering and binning, disintegration), the matrix-convexity lemma, the uniqueness proposition, and the binary theorem's symmetrization and positivity arguments are verified by written proof and the numerical harnesses, not by the proof assistant. The scripts, their archived run outputs, and the full check inventory accompany the manuscript sources at the repository of the title footnote (`VERIFICATION.md` in the paper directory), together with an executed notebook, `reproduce.ipynb`, submitted as supplementary material, which evaluates every closed form against direct minimization, regenerates the figures and tables, and runs the three harnesses in one pass.

APPENDIX C COMPARISON WITH THE BINARY COMMON-RECONSTRUCTION FUNCTION OF LU *et al.*

Lu, Xu, Zhang, Feng, and Wang [21] solve the binary lossy coding problem with encoder side information and a common-reconstruction constraint; its Heegard–Berger companion is [22]. Their model has a source X , encoder side information S_1 , and decoder side information S_2 forming $X - S_1 - S_2$, with the encoder observing (X, S_1) , the decoder reconstructing from the message and S_2 , and the common-reconstruction constraint that the encoder reproduce the decoder's estimate. For the doubly symmetric binary source with $\Pr[X \neq S_1] = p_1$ and $S_2 = S_1 \oplus \text{Bern}(p_2)$ under Hamming distortion, their Theorem 1 gives the achievable rate

$$\tilde{R}^{\text{CR}}(D) = \min_{W: W - (X, S_1) - S_2, \mathbb{E}d(X, f(W)) \leq D} I(X, S_1; W | S_2),$$

evaluated in closed form in their Theorem 2.

Under the relabeling

$$X \leftrightarrow Y, \quad S_1 \leftrightarrow V, \quad S_2 \leftrightarrow S, \quad W \leftrightarrow \hat{Y}, \quad (p_1, p_2) \leftrightarrow (p, q),$$

their functional is identically the conditional-content functional of Section VII: $I(X, S_1; W | S_2) = I(Y, V; \hat{Y} | S)$ under the same Markov chain $\hat{Y} - (Y, V) - S$ and the same Hamming constraint, which by pair sufficiency (Lemma 11) is L . The binary conditional-content function $L(D)$ of this paper therefore *coincides* with $\tilde{R}^{\text{CR}}(D)$, and we do not claim it as new. The endpoints agree accordingly: at $q = 0$ ($p_2 = 0$) both reduce to Gray's conditional function $h_2(p) - h_2(D)$ (their Remark 3), the $q = 0$ anchor of Theorem 27; at $q = \frac{1}{2}$ ($p_2 = \frac{1}{2}$) both reduce to the marginal rate-distortion function $1 - h_2(D)$; and at $D = 0$ both give $h_2(p * q)$ (their Remark 1).

The operational readings differ, and the second coordinate is new. In [21] the conditioning variable S_2 is decoder side information, so $I(X, S_1; W | S_2)$ is a transmitted *rate* with a Wyner–Ziv saving; here S is held by a third party that never decodes, so the same quantity is the conditional *content* L , while the transmitted rate is instead the unconditional $R = I(Y, V; \hat{Y})$, which their single-coordinate problem does not price. The joint (R, L) region and the tilt parametrization of the whole frontier (Proposition 28) have no counterpart in [21]: they price both coordinates of one description at once, rather than the single common-reconstruction rate, and are the binary contribution of this paper.

REFERENCES

- [1] C. E. Shannon, "Coding theorems for a discrete source with a fidelity criterion," in *IRE Nat. Conv. Rec.*, 1959, pp. 142–163.
- [2] T. Berger, *Rate Distortion Theory: A Mathematical Basis for Data Compression*. Englewood Cliffs, NJ: Prentice-Hall, 1971.
- [3] R. M. Gray, "Conditional rate-distortion theory," Stanford Electronics Laboratories, Stanford, CA, Tech. Rep. 6502-2, Oct. 1972.
- [4] R. M. Gray, "A new class of lower bounds to information rates of stationary sources via conditional rate-distortion functions," *IEEE Trans. Inf. Theory*, vol. 19, no. 4, pp. 480–489, 1973.
- [5] A. D. Wyner and J. Ziv, "The rate-distortion function for source coding with side information at the decoder," *IEEE Trans. Inf. Theory*, vol. 22, no. 1, pp. 1–10, 1976.
- [6] M. S. Pinsker, *Information and Information Stability of Random Variables and Processes*. San Francisco, CA: Holden-Day, 1964.
- [7] E. C. Posner, "Random coding strategies for minimum entropy," *IEEE Trans. Inf. Theory*, vol. 21, no. 4, pp. 388–391, 1975.
- [8] H. Cramér, "Über eine Eigenschaft der normalen Verteilungsfunktion," *Math. Z.*, vol. 41, no. 1, pp. 405–414, 1936.
- [9] A. H. Kaspí, "Rate-distortion function when side-information may be present at the decoder," *IEEE Trans. Inf. Theory*, vol. 40, no. 6, pp. 2031–2034, 1994.
- [10] C. Heegard and T. Berger, "Rate distortion when side information may be absent," *IEEE Trans. Inf. Theory*, vol. 31, no. 6, pp. 727–734, 1985.
- [11] Y. Steinberg, "Coding and common reconstruction," *IEEE Trans. Inf. Theory*, vol. 55, no. 11, pp. 4995–5010, 2009.
- [12] A. Lapidoth, A. Malär, and M. Wigger, "Constrained source-coding with side information," *IEEE Trans. Inf. Theory*, vol. 60, no. 6, pp. 3218–3237, 2014.
- [13] B. Ahmadi, R. Tandon, O. Simeone, and H. V. Poor, "Heegard-Berger and cascade source coding problems with common reconstruction constraints," *IEEE Trans. Inf. Theory*, vol. 59, no. 3, pp. 1458–1474, 2013.
- [14] H. Yamamoto, "Rate-distortion theory for the Shannon cipher system," *IEEE Trans. Inf. Theory*, vol. 43, no. 3, pp. 827–835, 1997.
- [15] H. Yamamoto, "A source coding problem for sources with additional outputs to keep secret from the receiver or wiretappers," *IEEE Trans. Inf. Theory*, vol. 29, no. 6, pp. 918–923, 1983.
- [16] J. Villard and P. Piantanida, "Secure multiterminal source coding with side information at the eavesdropper," *IEEE Trans. Inf. Theory*, vol. 59, no. 6, pp. 3668–3692, 2013.
- [17] C. Schieler and P. Cuff, "Rate-distortion theory for secrecy systems," *IEEE Trans. Inf. Theory*, vol. 60, no. 12, pp. 7584–7605, 2014.
- [18] R. Tandon, L. Sankar, and H. V. Poor, "Discriminatory lossy source coding: Side information privacy," *IEEE Trans. Inf. Theory*, vol. 59, no. 9, pp. 5665–5677, 2013.
- [19] A. Makhdoomi, S. Salamatian, N. Fawaz, and M. Médard, "From the information bottleneck to the privacy funnel," in *Proc. IEEE Inf. Theory Workshop (ITW)*, 2014, pp. 501–505.
- [20] J. Armstrong, "Source-side sufficiency for the information bottleneck: Exact reduction and finite-block equivalence," arXiv:2604.26744, 2026.
- [21] J. Lu, Y. Xu, P. Zhang, M. Feng, and Q. Wang, "Binary lossy coding problem with encoder side information and common reconstruction constraint," in *Proc. Int. Conf. Wireless Commun. Signal Process. (WCSP)*, 2016, pp. 1–5.
- [22] J. Lu, Y. Xu, Y. Qian, and Q. Wang, "The binary Heegard-Berger problem with encoder side information and common reconstruction constraints," in *Proc. Int. Conf. Wireless Commun. Signal Process. (WCSP)*, 2016, pp. 1–5.
- [23] J. Lu and Y. Xu, "Vector Gaussian Heegard-Berger problem with common reconstructions," in *Proc. Int. Conf. Wireless Commun. Signal Process. (WCSP)*, 2021, pp. 1–5.
- [24] M. Meidlinger, A. Winkelbauer, and G. Matz, "On the relation between the Gaussian information bottleneck and MSE-optimal rate-distortion quantization," in *Proc. IEEE Workshop Statistical Signal Process. (SSP)*, 2014, pp. 89–92.
- [25] G. Chechik, A. Globerson, N. Tishby, and Y. Weiss, "Information bottleneck for Gaussian variables," *J. Mach. Learn. Res.*, vol. 6, pp. 165–188, 2005.
- [26] N. Tishby, F. C. Pereira, and W. Bialek, "The information bottleneck method," in *Proc. 37th Allerton Conf. Commun., Control, Comput.*, Monticello, IL, 1999, pp. 368–377.

- [27] R. Landauer, "Irreversibility and heat generation in the computing process," *IBM J. Res. Dev.*, vol. 5, no. 3, pp. 183–191, 1961.
- [28] C. H. Bennett, "Logical reversibility of computation," *IBM J. Res. Dev.*, vol. 17, no. 6, pp. 525–532, 1973.
- [29] L. del Rio, J. Åberg, R. Renner, O. Dahlsten, and V. Vedral, "The thermodynamic meaning of negative entropy," *Nature*, vol. 474, pp. 61–63, 2011.
- [30] P. Faist, F. Dupuis, J. Oppenheim, and R. Renner, "The minimal work cost of information processing," *Nature Commun.*, vol. 6, art. 7669, 2015.
- [31] M. Esposito and C. Van den Broeck, "Second law and Landauer principle far from equilibrium," *Europhys. Lett.*, vol. 95, art. 40004, 2011.
- [32] T. Sagawa and M. Ueda, "Fluctuation theorem with information exchange: Role of correlations in stochastic thermodynamics," *Phys. Rev. Lett.*, vol. 109, art. 180602, 2012.
- [33] T. Sagawa and M. Ueda, "Minimal energy cost for thermodynamic information processing: Measurement and information erasure," *Phys. Rev. Lett.*, vol. 102, art. 250602, 2009.
- [34] J. M. R. Parrondo, J. M. Horowitz, and T. Sagawa, "Thermodynamics of information," *Nature Phys.*, vol. 11, pp. 131–139, 2015.
- [35] A. H. Bond, "A rate–work–distortion region for consumer-relative observation," preliminary manuscript, 2026, archived at doi:10.5281/zenodo.21776291.
- [36] S. Boyd and L. Vandenberghe, *Convex Optimization*. Cambridge, U.K.: Cambridge Univ. Press, 2004.
- [37] I. Csiszár, "Information-type measures of difference of probability distributions and indirect observations," *Studia Sci. Math. Hungar.*, vol. 2, pp. 299–318, 1967.

Andrew H. Bond received the B.S.E.E. degree from North Carolina State University and the M.S.S.E. degree from San José State University. He has over 20 years of industry experience at AT&T, Cisco Systems, and Fujitsu Network Communications. He is currently with the Department of Computer Engineering, San José State University, San José, CA, USA, where he teaches and advises graduate theses. His research interests include computational models of social and economic behavior, embedding compression, autonomous security agents, and cognitive architectures. He is a Senior Member of IEEE.